BSP Build Guide

for Windows®CE 5.0

MMSP™2

Build Guide

MDMP2520FBBUM-WC50

Ver. 5.4.2 / September 15, 2005

MP2520F

Platform Builder for Microsoft Windows® CE 5.0

for MMSP 2 Application Processor

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I N D E X

WinCE.NET BSP Build Guide for MP2520F DTK4

1. Creating a New Platform

1.1 Installing BSP

Create **MMSP20** folder under **WINCE500\PLATFORM** folder

Uncompress BSP File in CD to **WINCE500\PLATFORM\MMSP20** Folder

CAUTION

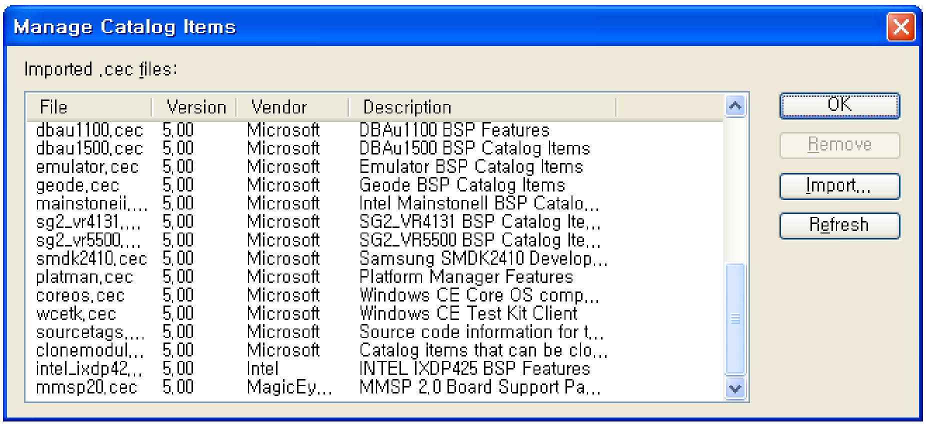
- If you have previous version of the BSP, remove it completely before install the new version of the BSP..
- Close platform builder and editor application before you install the new version of BSP.

1.2 Catalog Registration of BSP

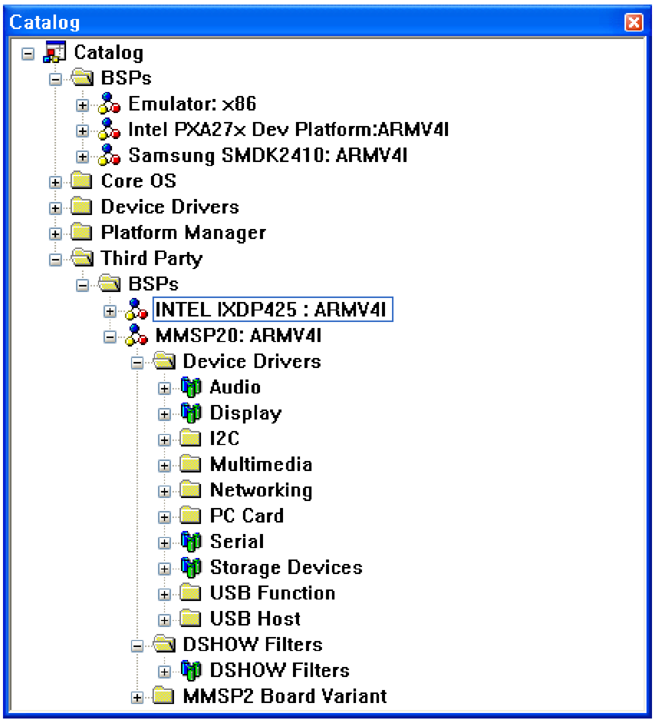
1. Run Platform Builder

2. Select **Manage Catalog Features** in **File Menu**..

3. Press **Import** button and select **MMSP20.cec** in MMPS20 folder.

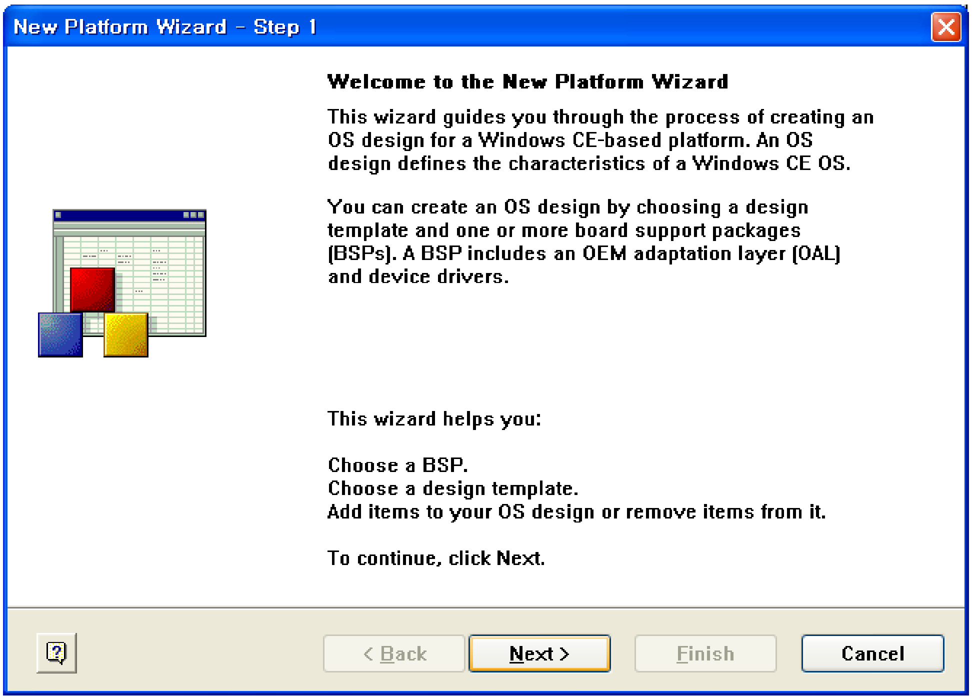


4. Register imported .cec file by pressing OK.
5. Now MP2520F BSP is added in catalog list.

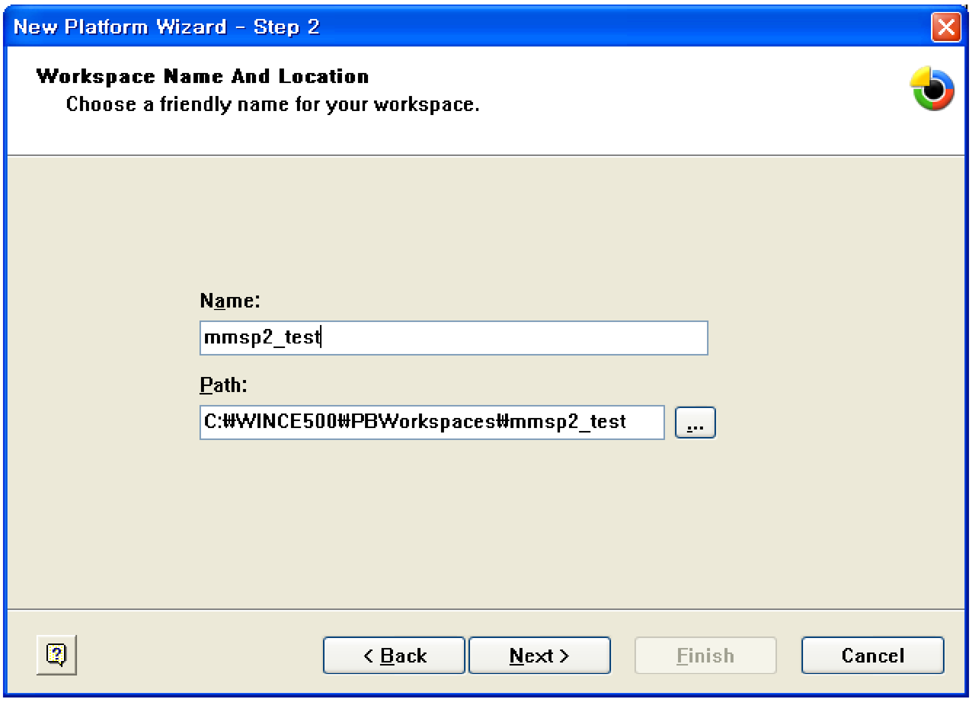

Catalog
Catalog
BSPs
Emulator: x86
Intel PXA27x Dev Platform:ARMV4I
Samsung SMDK2410: ARMV4I
Core OS
Device Drivers
Platform Manager
Third Party
BSPs
INTEL IXDP425 : ARMV4I
MMSP20: ARMV4I
Device Drivers
Audio
Display
I2C
Multimedia
Networking
PC Card
Serial
Storage Devices
USB Function
USB Host
DSHOW Filters
DSHOW Filters
MMSP2 Board Variant

1.3 Creating New MMSP20 Platform

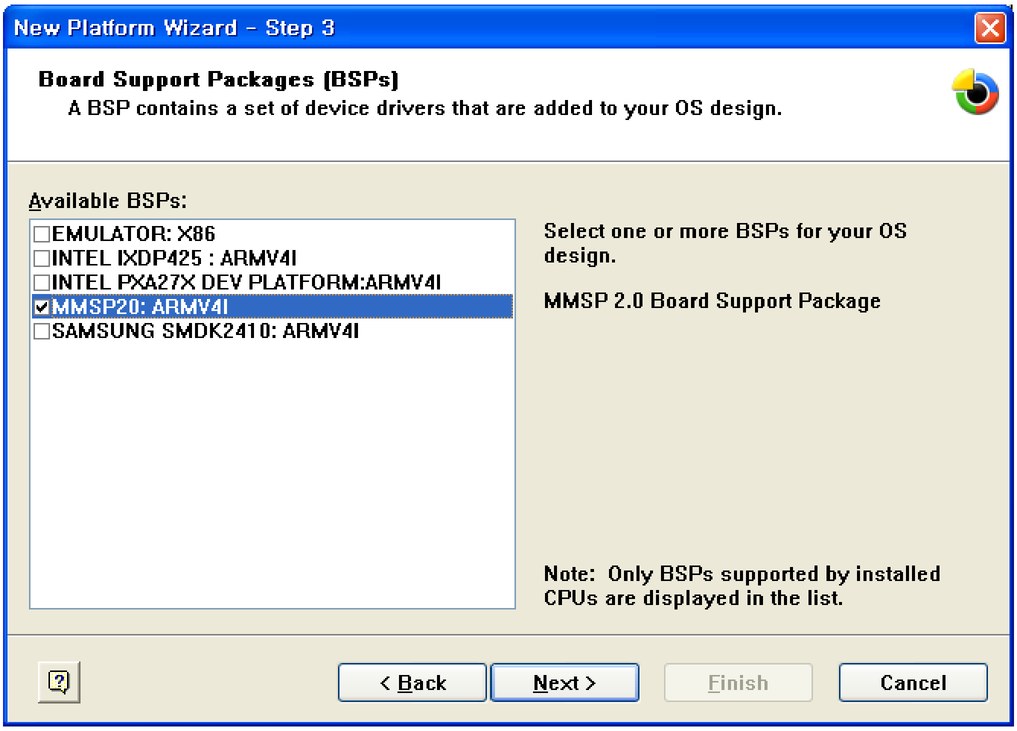
1. Go to **File** -> **New Platform** in Platform Builder menu.



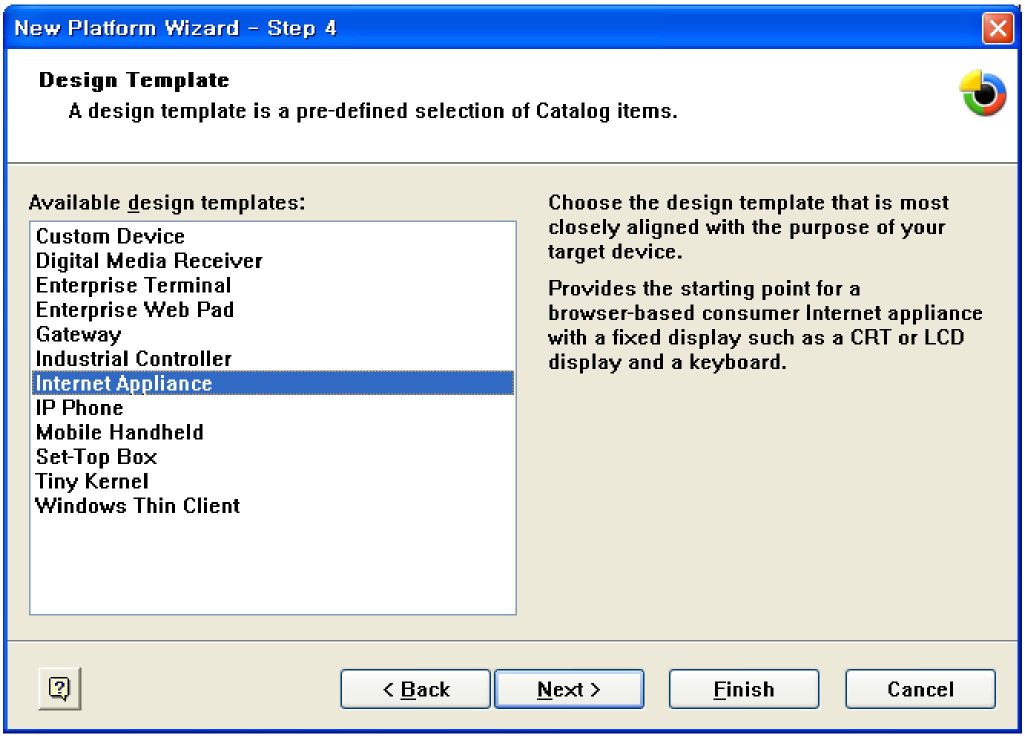
2. Press **Next** and choose a name for new workplace.



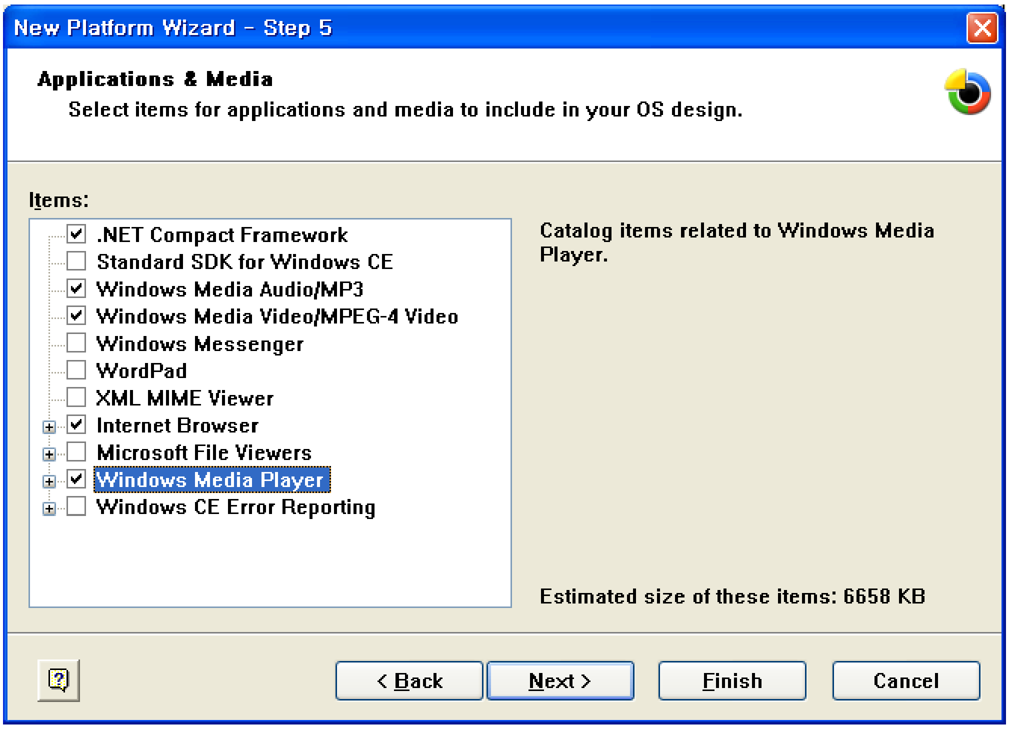
3. Press **Next** and check **MMSP20:ARMV4I** in New Platform Wizard.



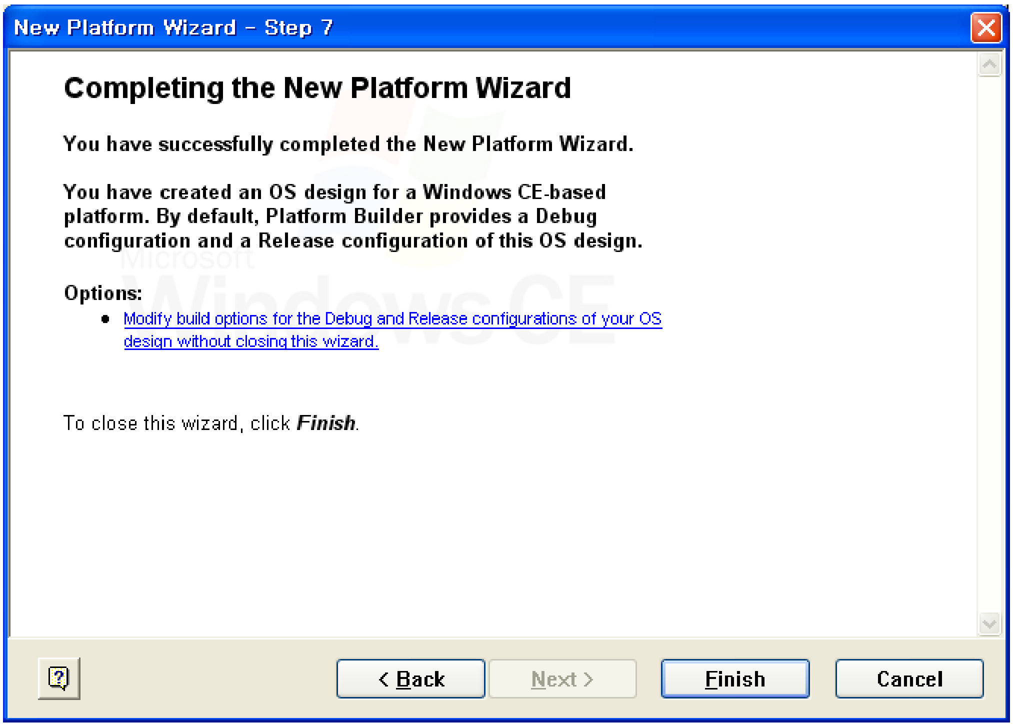
4. Select appropriate configuration template.



5. Press **Next** > and check features needed. (Features shown in this window can be added or removed in Catalog window.)



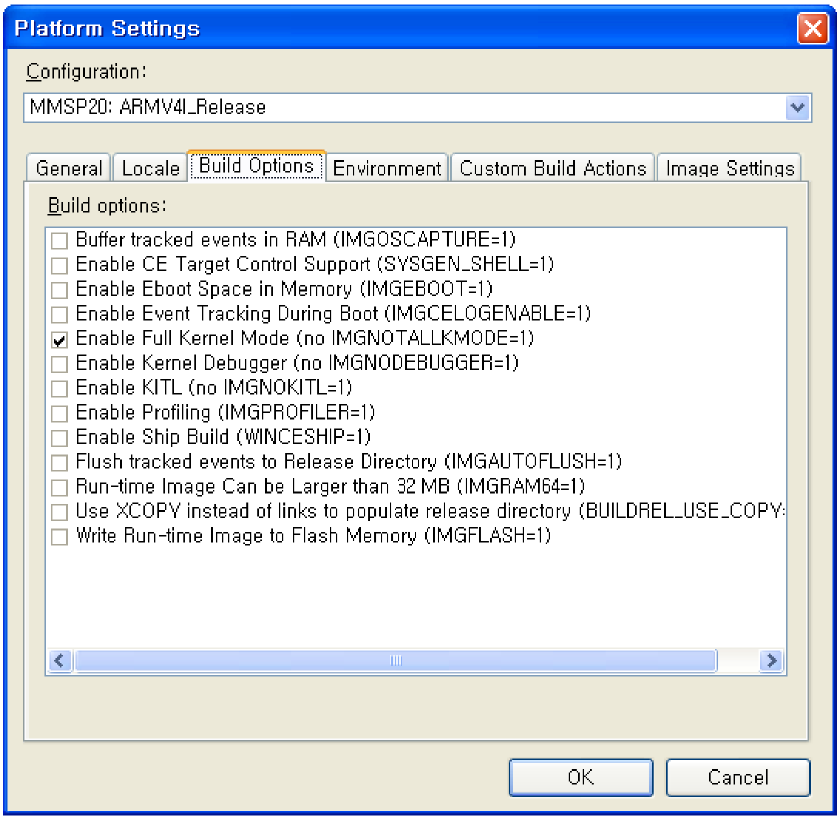
6. Complete New Platform creation by pressing **Finish** button.



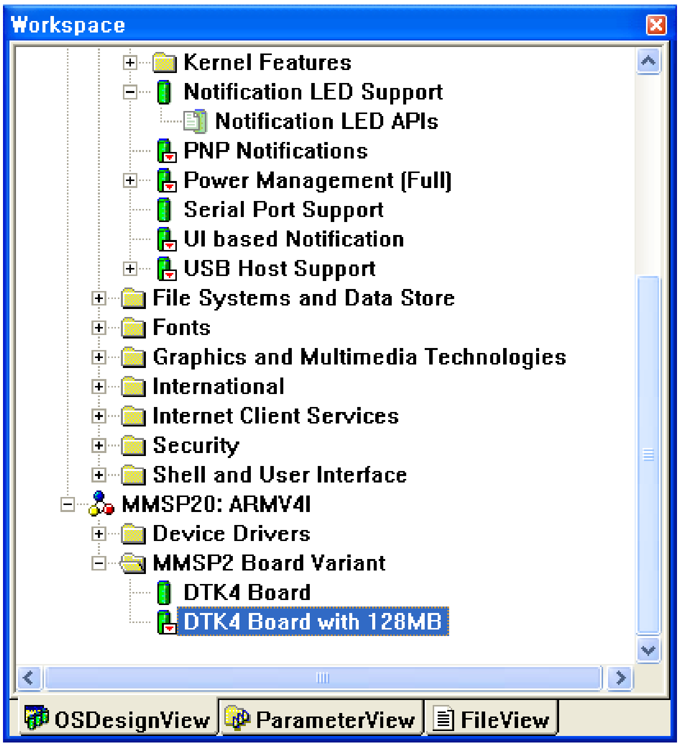
2. Platform Configuration

2.1 Configuration Settings for Development Board

1. Select **Platform** -> **Settings** in Platform Builder Menu.
2. Check **Enable Full Kernel Mode** in the **Build Options** list.



Add appropriate board configuration from Catalog Window.

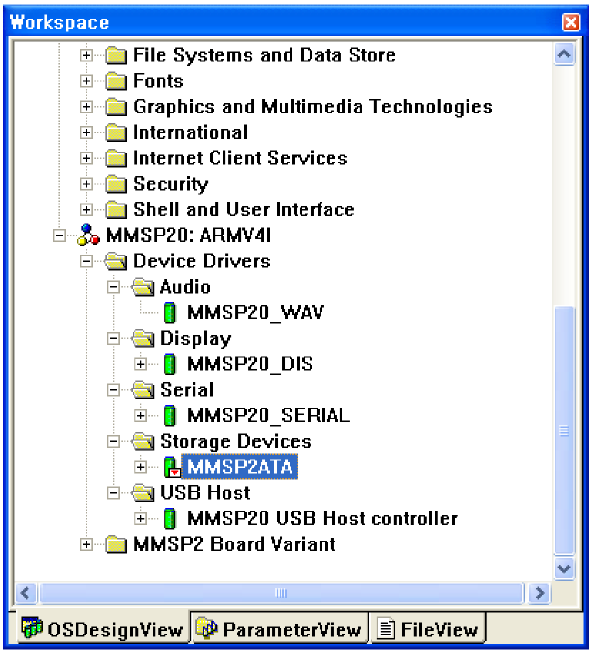


DTK4 Board: Configuration for DTK 4^{th} board. This configuration sets BSP_DTK4 environment variable and enables "#define PMP_BOARD" in the BSP source code.

DTK4 Board with 128MB: This enables memory settings for 128MB DRAM configuration.

3. Set DIP switch on the board appropriately. (See "Output Mode Selection" of the Display Driver Guide)

2.2 Adding Basic Components



Add necessary components according to the driver guides in this document.

(**Errors can be occurred during build process when components required by each driver are not added and configured correctly. For example, Display Driver is built correctly when DirectDraw component is included.**)

For configuring the drivers and components, you must refer to each guide in chapter 3 through 11.

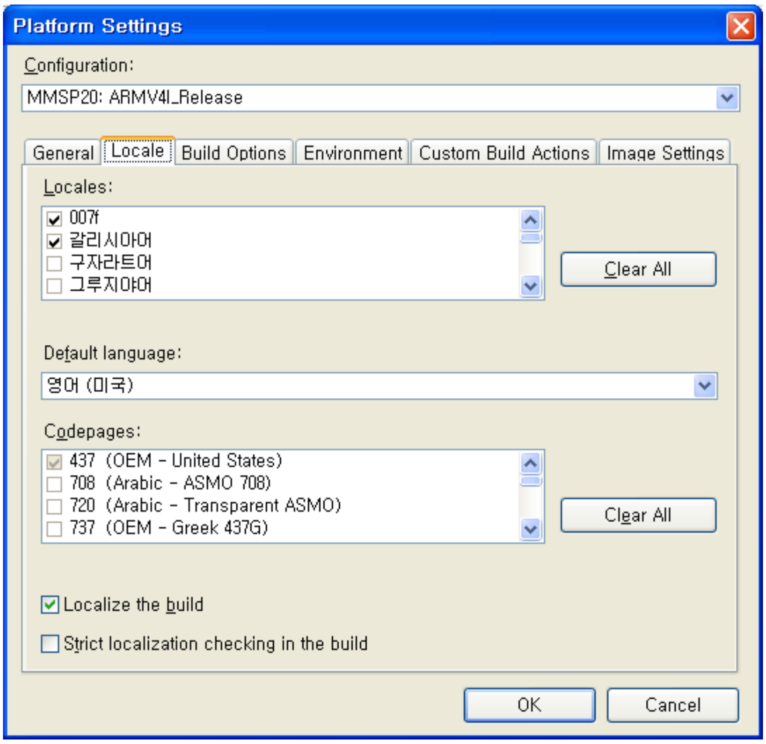
You can reduce image of the Windows CE OS by changing default language to English rather than Asian language such as Korean, Chinese, or Janapense. To change default language:

Select **Platform** -> **Settings** Menu.

Clear all locales by pressing **Clear All** button of **Locales:**.

Select English from **Locales:** list.

Select **Default Language** as English.



This is a sample of locale setting of Platform Builder Workspace than runs on Korean Windows XP.

3. Display Driver Guide

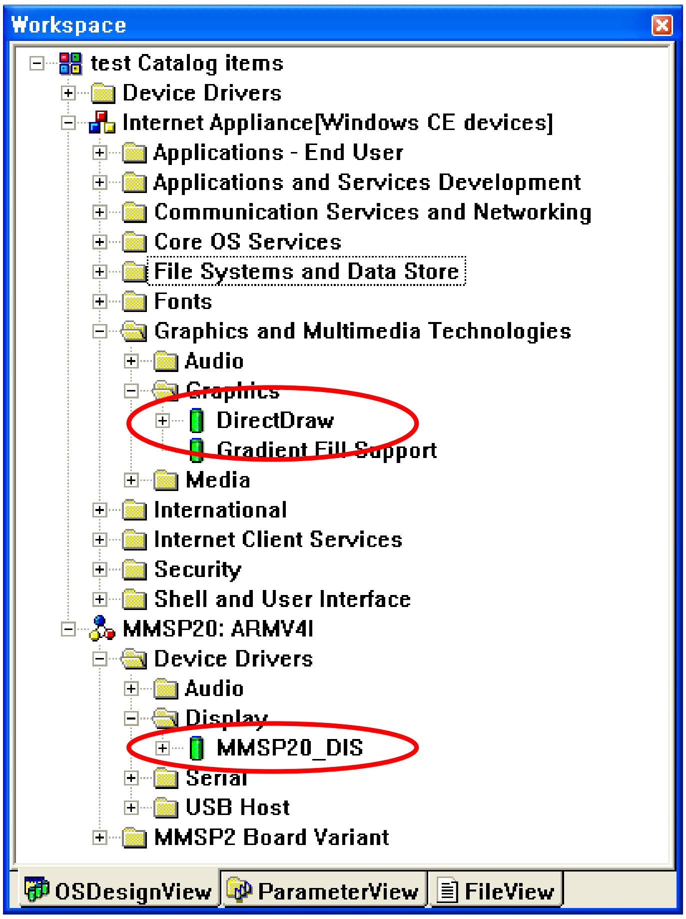
3.1 Adding Display Driver

3.1.1 Adding Display device driver

Add **MMSP20_DIS** component in **Third Party** → **BSPs** → **MMSP20: ARMV4I** → **Device Drivers** →**Display** of the catalog. (Click this item with right button of the mouse and select 'Add to OS Design' menu.)

3.1.2 Adding DirectDraw component

Add **DirectDraw** component in **Core OS** → **Windows CE Devices** → **Graphics and Multimedia Technologies** → **Graphics** of the catalog



3.2 Output Mode Selection

The development has a array of DIP switch and it is used for display output mode selection.

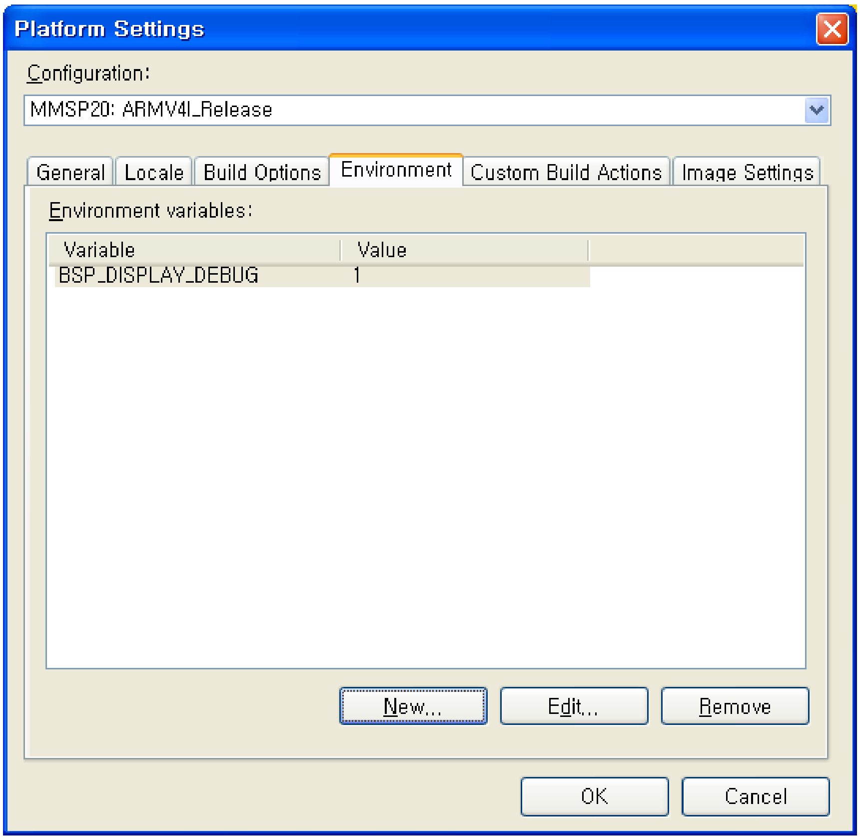
DIP switch settings for DTK revision 4.11 is shown below:

Connector	Resolution	SW2 (DIP Switch)	Remark

		4	3	2	1	
LCD	800x480x60 Hz	-	OFF	OFF	OFF	Default setting for DTK 4.11
D-SUB (Monitor)	640x480x60 Hz	-	OFF	OFF	ON	CX25874
D-SUB (Monitor)	800x600x60 Hz	-	OFF	ON	OFF	CX25874
Component(YPbPr)	720x480px60 Hz	-	OFF	ON	ON	CX25874
-	-	-	ON	OFF	OFF	
Composite/S-Video	720x480ix30 Hz	-	ON	OFF	ON	CX25874
LCD	800x600x60 Hz	-	ON	ON	OFF	
LCD	640x480x60 Hz	-	ON	ON	ON	

3.3 Debug Output

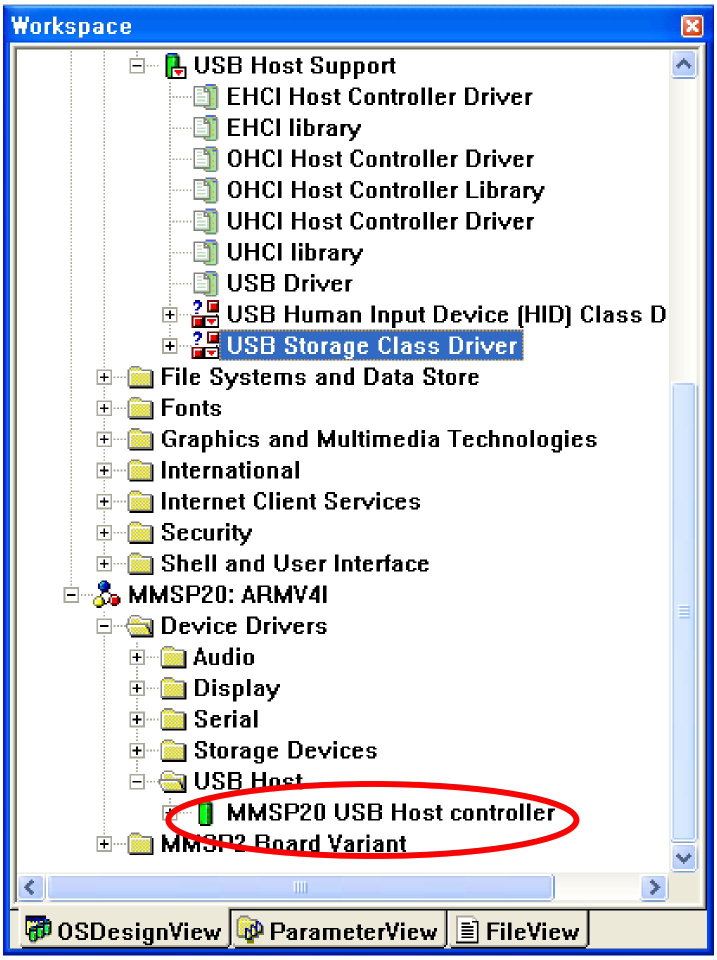
Detailed debugging message of the display driver can be shown in RETAIL mode when **BSP_DISPLAY_DEBUG** environment value is set to 1



4. USB Host Driver Guide

4.1 Adding USB Host Driver Component

Add **Third Party** → **BSPs** → **MMSP20:ARMV4I** → **Device Drivers** → **USB Host** → **USB Host Controllers** → **MMSP20 USB Host Controller** component. (Click this item with right button of the mouse and select 'Add to OS Design' menu.)



4.2 Using USB Mouse

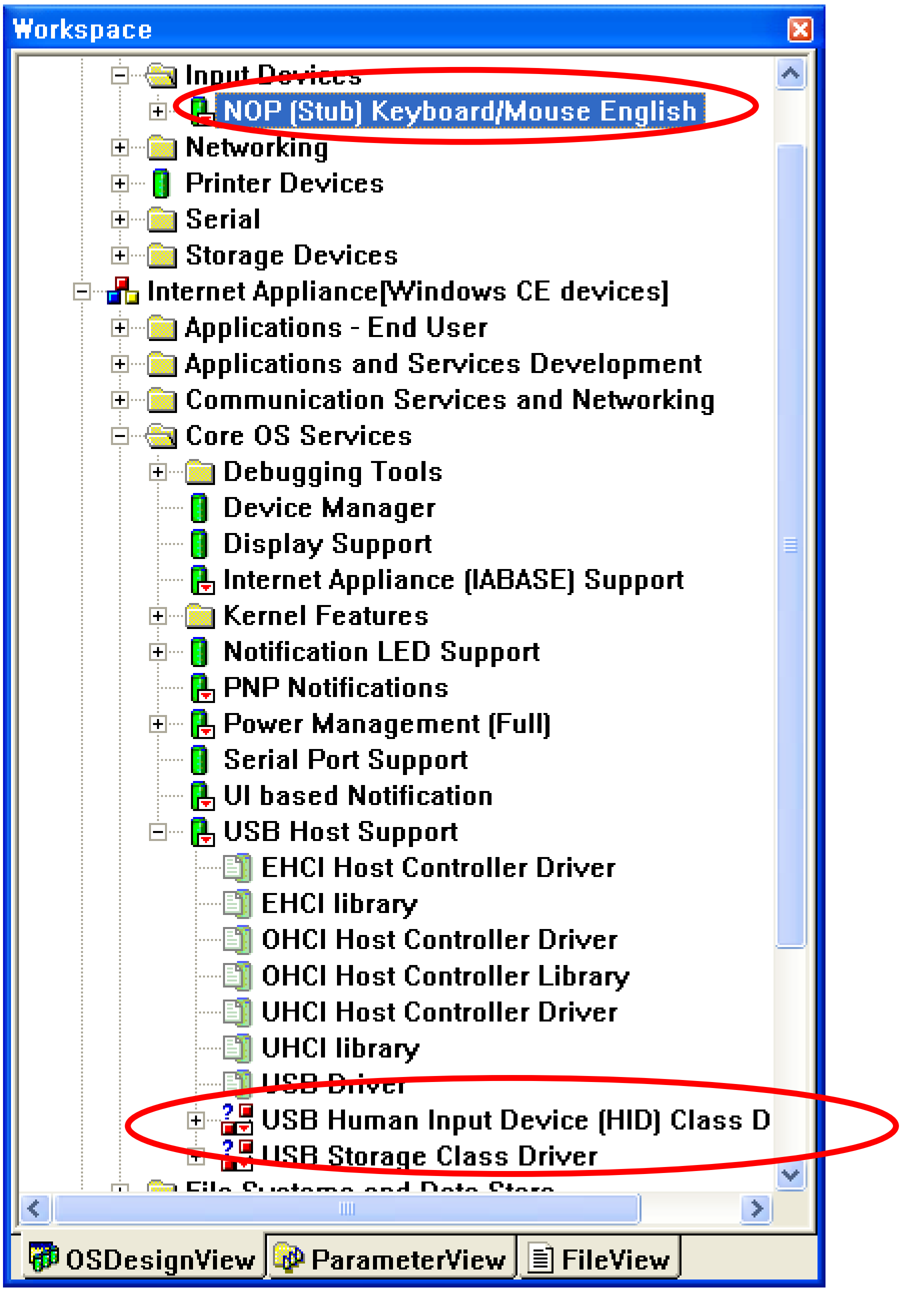
To use USB mouse, add **Core OS** → **Windows CE devices** → **Core OS Services** → **USB Host Support** → **USB Human Input Device (HID) Class Driver** → **USB HID Keyboard and Mouse** component. (See previous picture)

4.3 Using USB Keyboard

To use USB keyboard, add **USB HID Keyboard and Mouse** component and **Device Drivers -> Input Devices -> Keyboard/Mouse -> NOP(Stub) Keyboard/Mouse English** component.

4.4 Using USB Storage Device

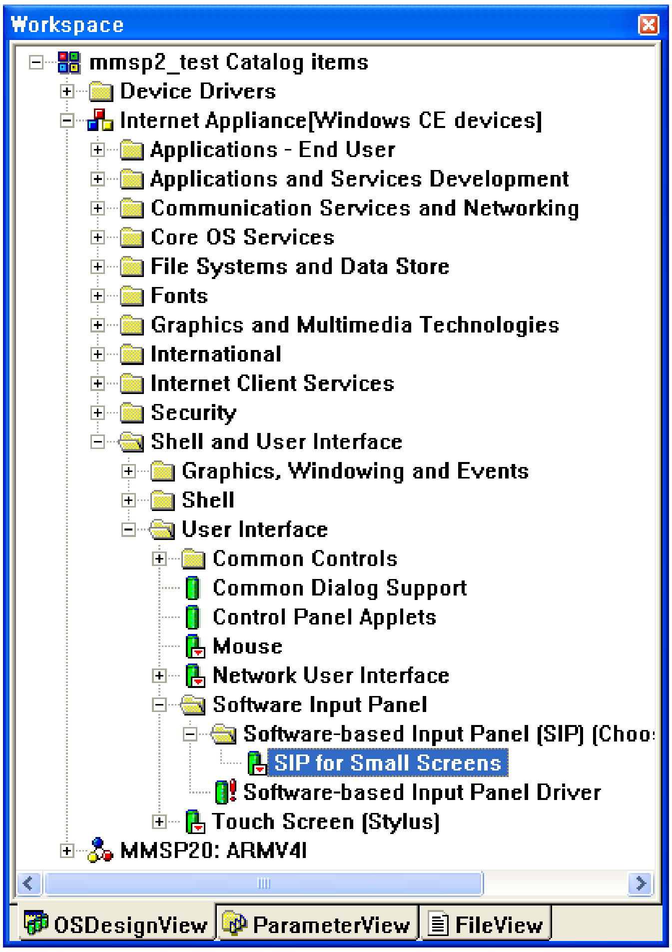
To use USB storage devices, add **Core OS** → **Windows CE Devices** → **Core OS Services** → **USB Host Support** → **USB Storage Class Driver** component. (See next picture)


Workspace
Input Devices
NOP (Stub) Keyboard/Mouse English
Networking
Printer Devices
Serial
Storage Devices
Internet Appliance[Windows CE devices]
Applications - End User
Applications and Services Development
Communication Services and Networking
Core OS Services
Debugging Tools
Device Manager
Display Support
Internet Appliance (IABASE) Support
Kernel Features
Notification LED Support
PNP Notifications
Power Management (Full)
Serial Port Support
UI based Notification
USB Host Support
EHCI Host Controller Driver
EHCI library
OHCI Host Controller Driver
OHCI Host Controller Library
UHCI Host Controller Driver
UHCI library
USB Driver
USB Human Input Device (HID) Class D
USB Storage Class Driver
OSDesignView
ParameterView
FileView

4.5 Using Virtual Keyboard

To use Virtual Keyboard instead of USB keyboard, **add Core OS → Windows CE Devices → Shell and User Interface → User Interface → Software Input Panel → Software-based Input Panel [SIP] [Choose 1 or more] → SIP for Large/Small Screens** component.

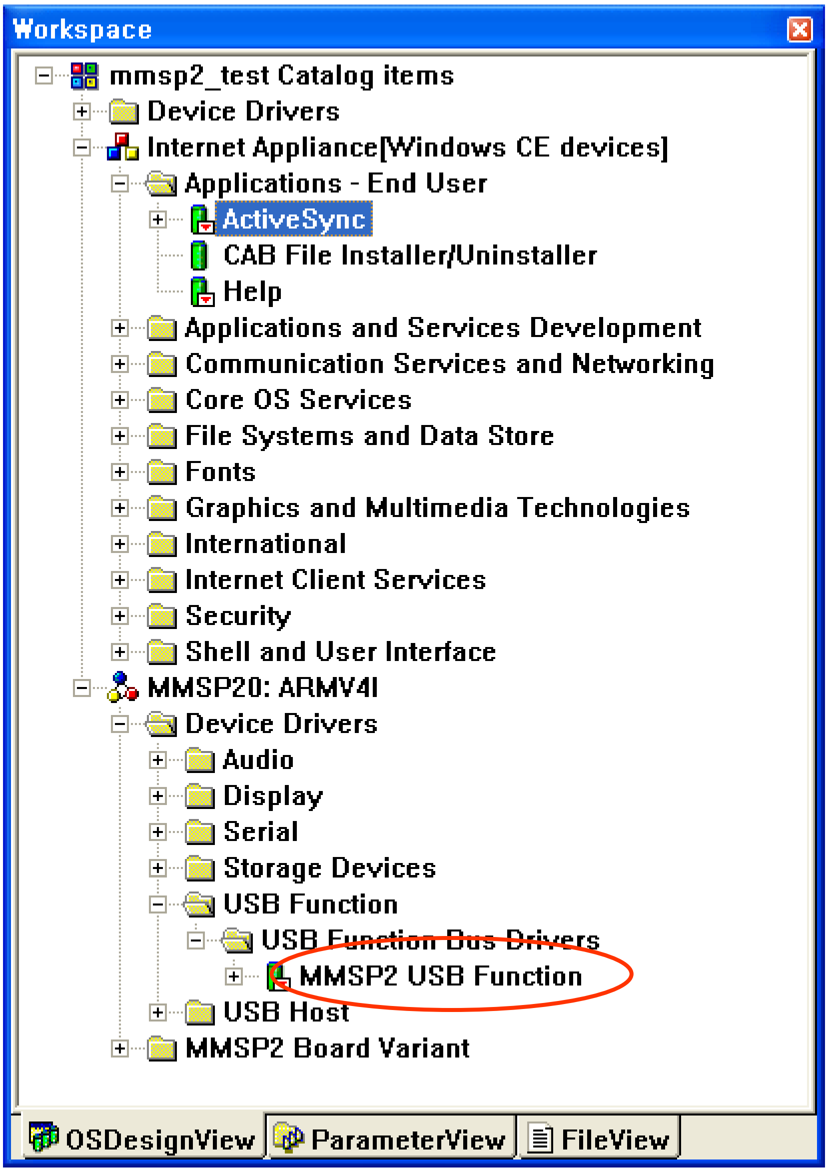
Windows CE supports two types of virtual keyboard, small and large. You can select either of them according to the screen configuration of your system.



5. USB 1.1 Function Driver Guide (on Chip)

5.1 Adding USB 1.1 Function Driver Component

Add **Third Party** → **BSPs** → **MMSP20:ARMV4I** → **Device Drivers** → **USB Function** → **USB Function Bus Drivers** → **MMSP2 USB Function** component. (Click this item with right button of the mouse and select 'Add to Platform' menu.)



5.2 Using ActiveSync with USB Connection

To test ActiveSync on DTK board, make sure things below are done.

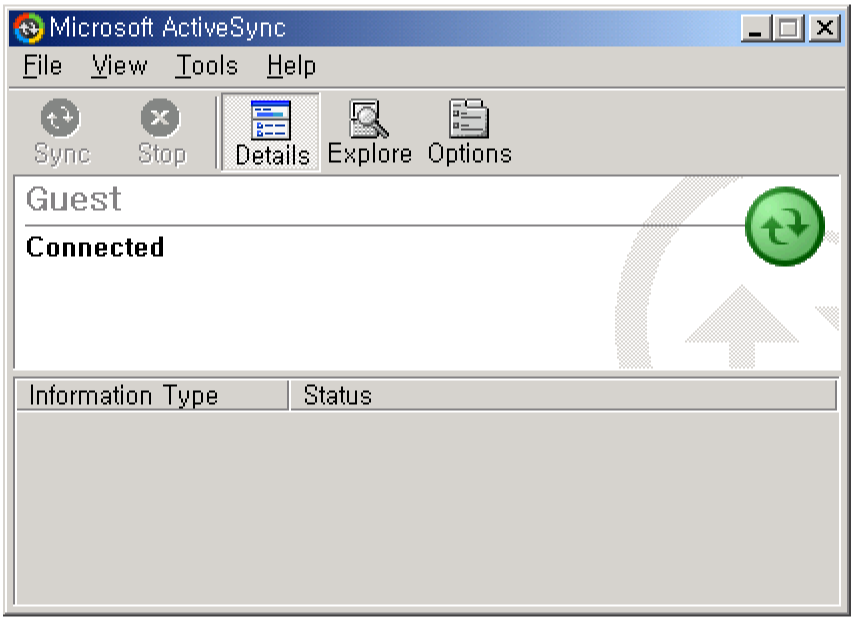
- USB connection cable is prepared. (One side is A type, and the other side B type)
- ActiveSync application is installed on host PC.
- When you set configuration of new project with **New Platform Wizard**, Check **ActiveSync** feature.
- Build image with above configuration and other settings shown on this BSP guide.

After Windows CE image is built. Follow steps shown below to establish ActiveSync connection with host PC.

1. Load Windows CE image on DTK board. (Do not connect USB cable yet)
2. Open **Control Panel** of the Windows CE.
3. Run **'Network and Dial-up Connections'** of Control Panel.
4. Select **'Make New Connection.'**
5. Select **'SC2410 USB Cable'** and press **'Finish'** button.
6. Run **'PC Connection'** of Control Panel.
7. Press **'Change...'** button.
8. Select 'My Connection.' (or other name you given when you created new connection)
9. Press OK button to exit.
10. Connect USB cable.
11. Try to remove and re-connect USB cable when ActiveSync connection is not established correctly.

NOTE: If you connect USB device for the first time, desktop Windows requires a device driver. The driver for USB device connection is in **DRIVERS\USB\FUNCTION** of the BSP source.

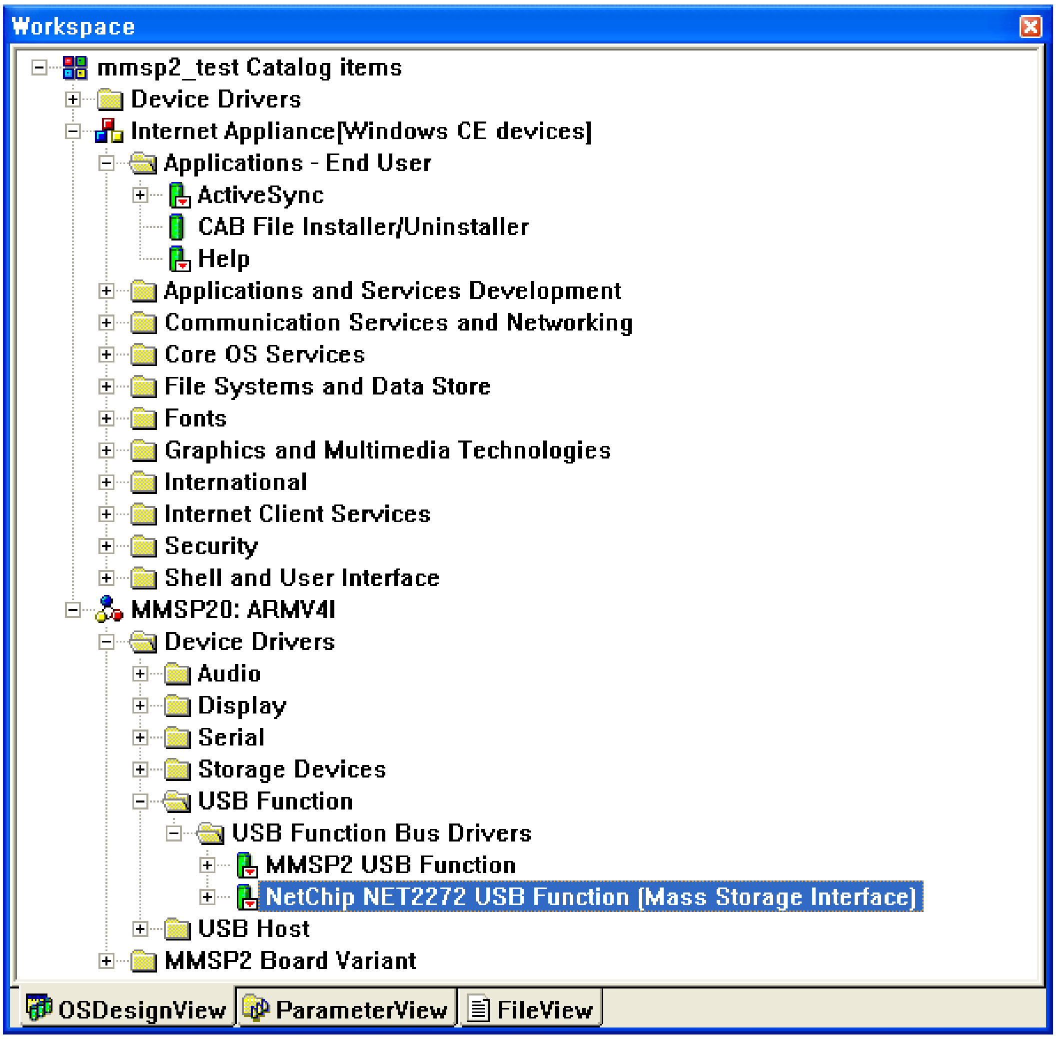
When some settings are completed, ActiveSync window will be shown like this:



6. USB 2.0 Mass Storage Driver Guide

6.1 Adding USB 2.0 Mass Storage Driver Component

Add **Third Party** → **BSPs** → **MMSP20:ARMV4I** → **Device Drivers** → **USB Function** → **USB Function Bus Drivers** →**NetChip NET2272 USB Function (Mass Storage Interface)** component. (Click this item with right button of the mouse and select 'Add to OS Design' menu.)



6.2 Connecting with Host

When USB 2.0 mass storage driver is initialized correctly, connect USB cable with host system. (e.g. desktop PC)

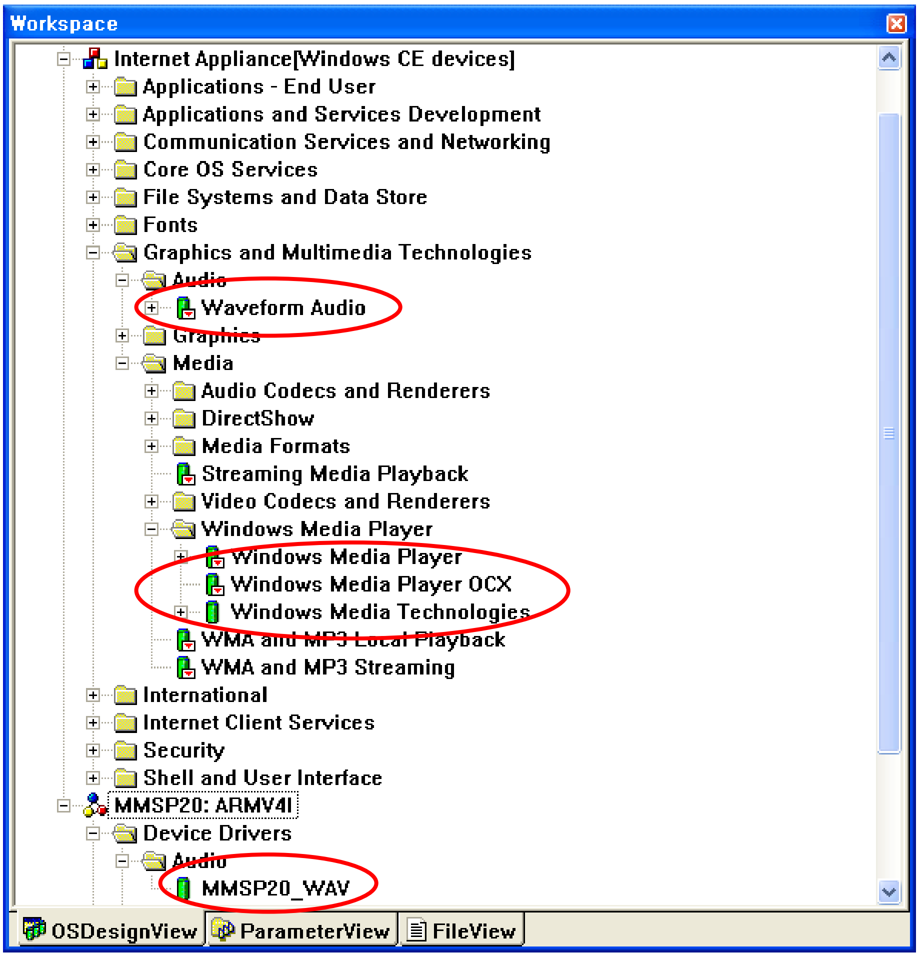
- On the DTK 4th board, use left one of the two USB device connectors.

After connection is established, MMSP2 based system is mounted as a removable disk on the host side.

7. AC97 Driver Guide

7.1 Adding Required Components

- Add **Third Party** → **BSPs** → **MMSP20:ARMV4I** → **Device Drivers** → **Audio** → **MMSP20_WAV** component.
- Add **Core OS** → **Windows CE Devices** → **Graphics and Multimedia Technologies** → **Audio** → **Waveform Audio** component for basic waveform audio play. (Click this item with right button of the mouse and select 'Add to OS Design' menu.)
- Add **Core OS** → **Windows CE Devices** → **Graphics and Multimedia Technologies** → **Media** → **Windows Media Player** → **Windows Media Player** as an audio player application. (make sure **Windows Media Player OCX** and **Windows Media Technologies** are also added)



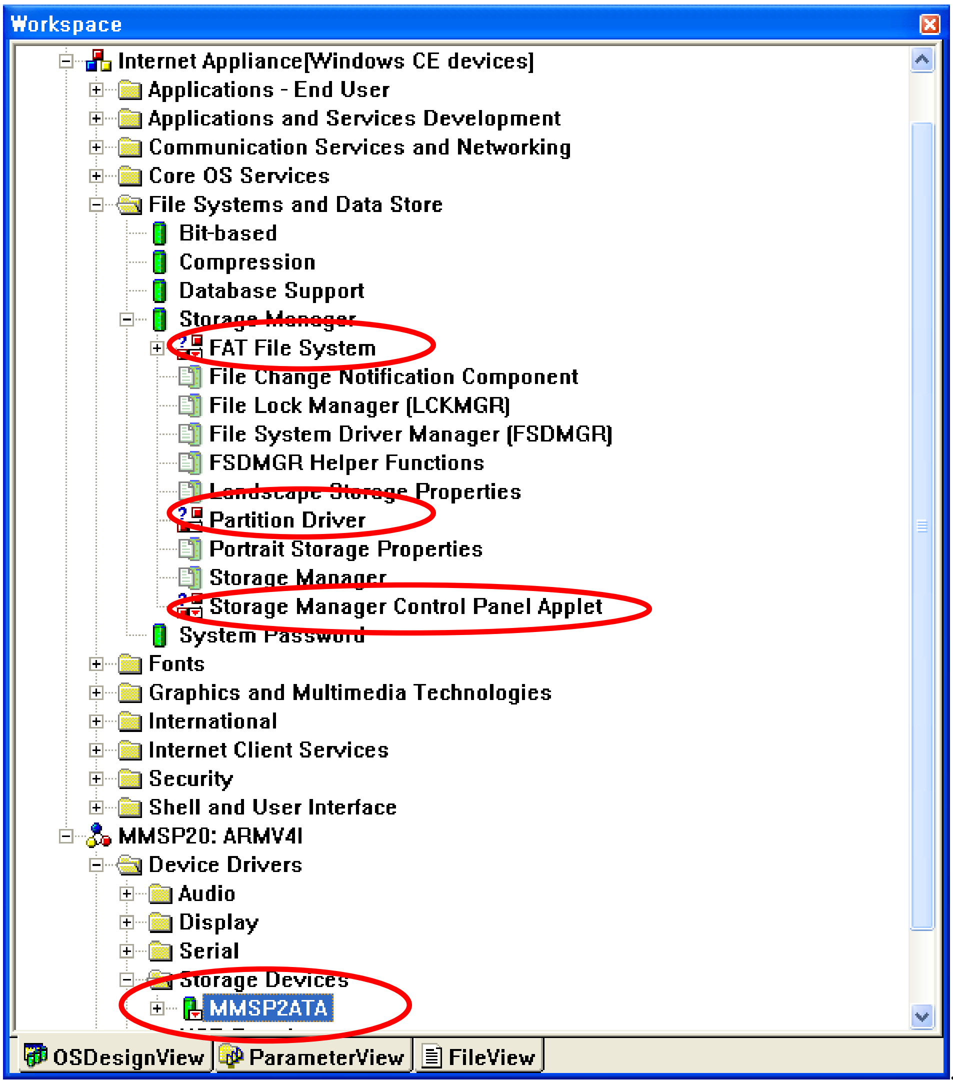
7.2 Adding Optional Components

To play other format of audio files, appropriate components should be added.

8. ATA Driver Guide

8.1 Adding ATA Device Driver

Add **Third Party** → **BSPs** → **MMSP20: ARMV4I** → **Device Drivers** → **Storage Devices** → **MMSP2ATA** component. (Click this item with right button of the mouse and select 'Add to OS Design' menu.)



8.2 Adding Required Components

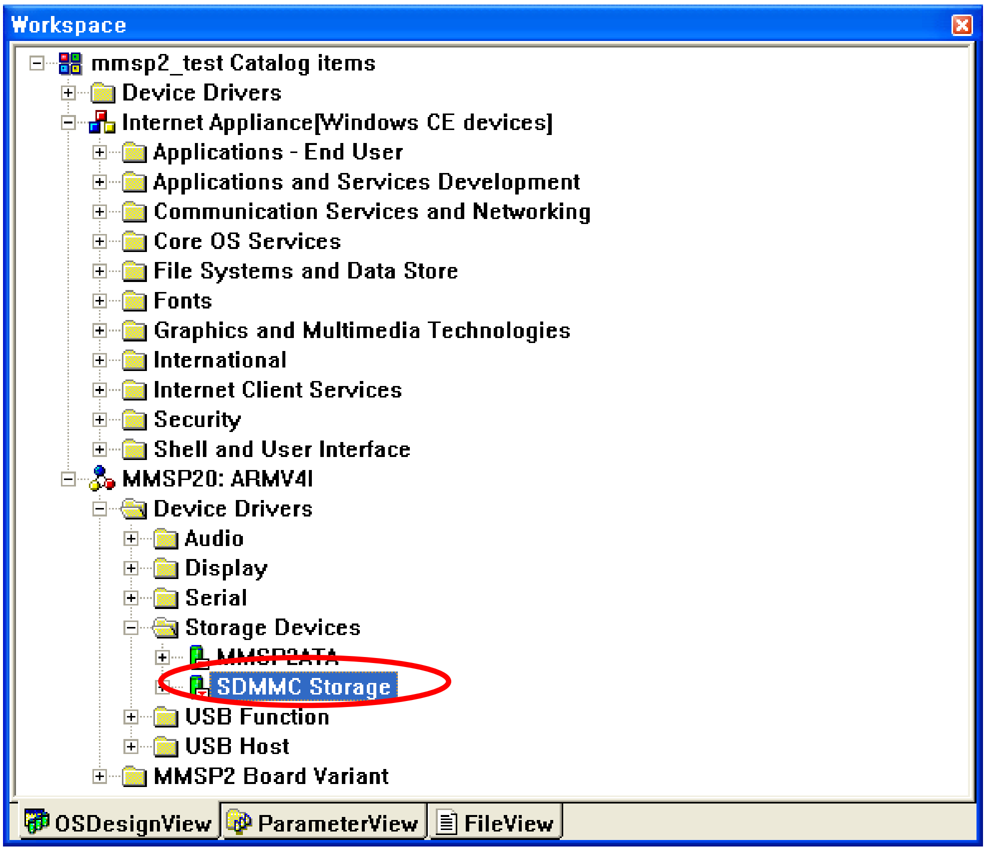
- Add below components in **Core OS** → **Windows CE Devices** → **File System and Data Store** → **Storage Manager**. (See above picture)

  - **FAT File System**
  - **Partition Driver**
  - **Storage Manager Control Panel Applet**

9. SD/MMC Storage Driver Guide

9.1 Adding SD/MMC Storage Device Driver

Add **Third Party** → **BSPs** → **MMSP20: ARMV4I** → **Device Drivers** → **Storage Devices** → **SDMMC Storage** component. (Click this item with right button of the mouse and select 'Add to Platform' menu.)



9.2 Features

- Supports Secure Digital (SD) Storage Card.

9.3 Limitations

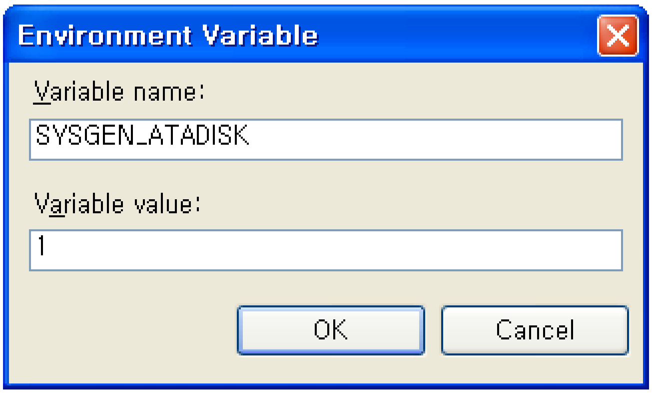
- Multimedia Card (MMC) is not supported.

9.4 Adding Required Components

- Add FAT File System Component so that disk can be formatted as FAT file system. (See 'Required Components' of the **ATA Drive** section)

10. CompactFlash Storage Driver Guide

You have to add a new environment variable, SYSGEN_ATADISK, into platform->settings and place it into TOP, not into ‘MMSP20:ARMV4I’.

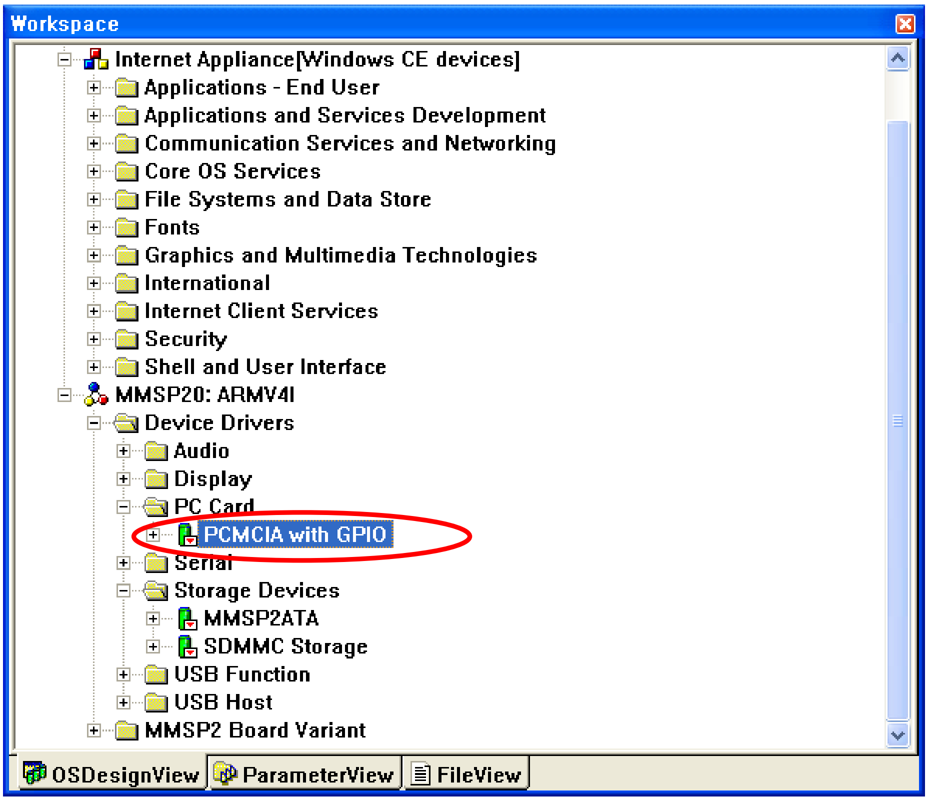


10.1 Adding CompactFlash Storage Device Driver

CompactFlash storage driver runs on the CompactFlash/PCMCIA bus driver layer. So you must include both CompcatFlash storage and CompactFlash/PCMCIA bus driver.

Add **Third Party** → **BSPs** → **MMSP20: ARMV4I** → **Device Drivers** → **PC Card** → **PC Card Host** → **PCMCIA with GPIO** component. (This is the CompactFlash / PCMCIA host driver. CF storage driver runs on this host layer.)

Add **Device Drivers** → **Storage Devices** → **Storage Devices** → **CompactFlash / PC Card Storage (ATADISK)** component. (This is the CompactFlash storage driver.)



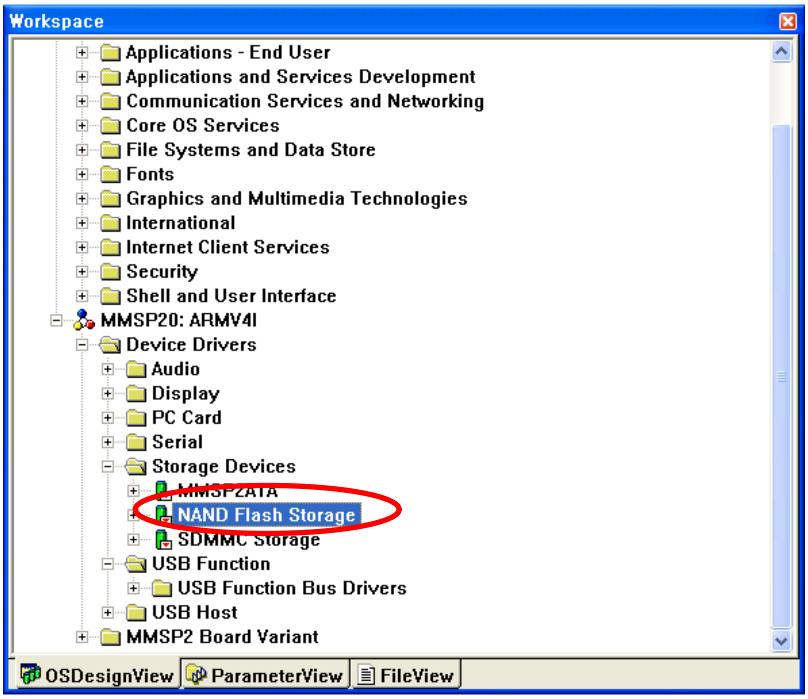
10.2 Adding Required Components

- Add FAT File System Component so that disk can be formatted as FAT file system. (See 'Required Components' of the **ATA Drive** section)

11. NAND Flash Storage Driver Guide

11.1 Adding NAND Flash Storage Device Driver

Add **Third Party** → **BSPs** → **MMSP20: ARMV4I** → **Device Drivers** → **Storage Devices** → **NAND Flash Storage** component.



11.2 Required Components

- Add FAT File System Component so that disk can be formatted as FAT file system. (See 'Required Components' of the **Chapter 7. ATA Drive** section)

11.3 Using Hive Registry with NAND Flash Storage

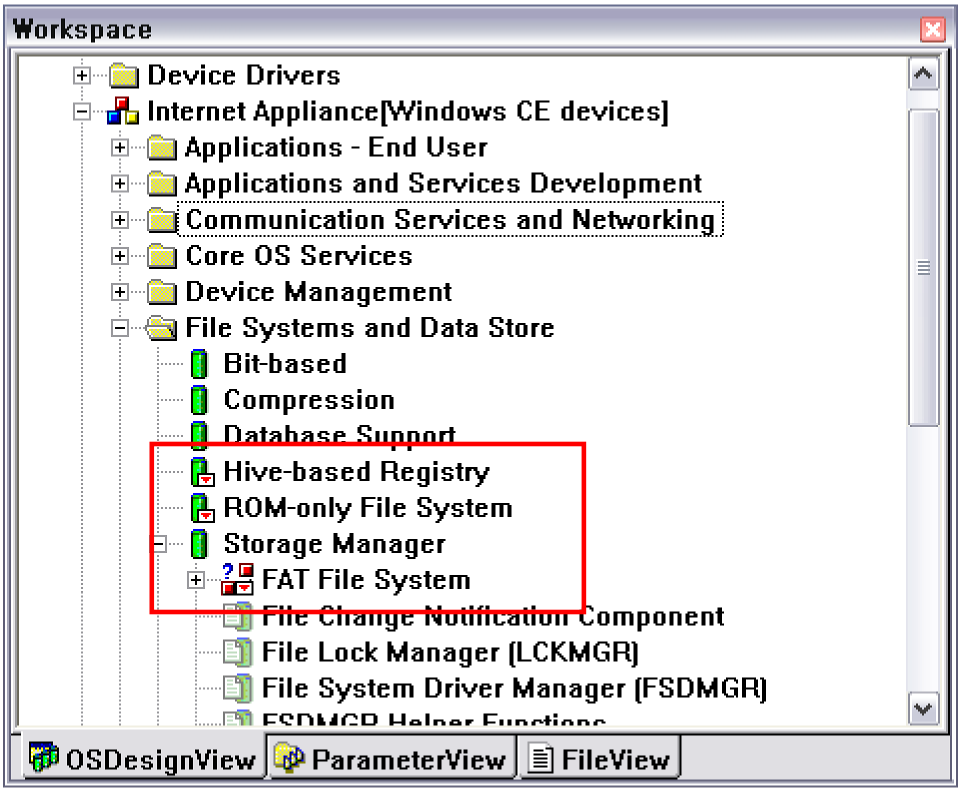
Windows CE OS image contains default registry settings. After OS is loaded, any modified registry value is stored in RAM. When Windows CE based system is reset, all modified registry value is modified.
To keep modified registry value during system reset, you can use Hive Registry feature of the Windows CE. With this feature, you can save new registry value in specified storage device.

CAUTION: We don't recommend hive enabled state as default setting. Use this functionality only when it fits to your purpose.

At first, add required features from catalog list.

- Core OS → Windows CE Devices → File Systems and Data Store → File System – Internal (Choose 1) → ROM-only File System
- Core OS → Windows CE Devices → File Systems and Data Store → Registry Store (Choose 1) → Hive-based Registry

And add all required components for NAND Flash Storage driver.



Then edit **platform.reg** file in the **FILES** folder. (You can also edit this file by selecting it on the **ParameterView** tab of the Platform Builder window.) Make sure registry values shown as bold are added.

```
; @CESYSGEN IF FILESYS_FSREGHIVE
; HIVE BOOT SECTION
[HKEY_LOCAL_MACHINE\init\BootVars]
   "SYSTEMHIVE"="Documents and Settings\\system.hv"
   "PROFILEDIR"="Documents and Settings"
   "Flags"=dword:3    ; Bit 0 - Storage Manager / Bit 1 - Device Manager
"RegistryFlags"=dword:0     ; 1 - Using aggressive flushing
; END HIVE BOOT SECTION
; @CESYSGEN ENDIF FILESYS_FSREGHIVE


; HIVE BOOT SECTION
IF BSP_FLASHDRV

[HKEY_LOCAL_MACHINE\Drivers\BuiltIn\FlashDrv]
       "Flags"=dword:1000

[HKEY_LOCAL_MACHINE\System\StorageManager\Profiles\FlashDrv]
    "BootPhase"=dword:0

 [HKEY_LOCAL_MACHINE\System\StorageManager\Profiles\FlashDrv\FATFS]
       "MountAsBootable"=dword:1
       "MountAsRoot"=dword:1

ENDIF
; END HIVE BOOT SECTION
```

This setting is applied after the OS is built by selecting **Build OS** -> **Sysgen**.

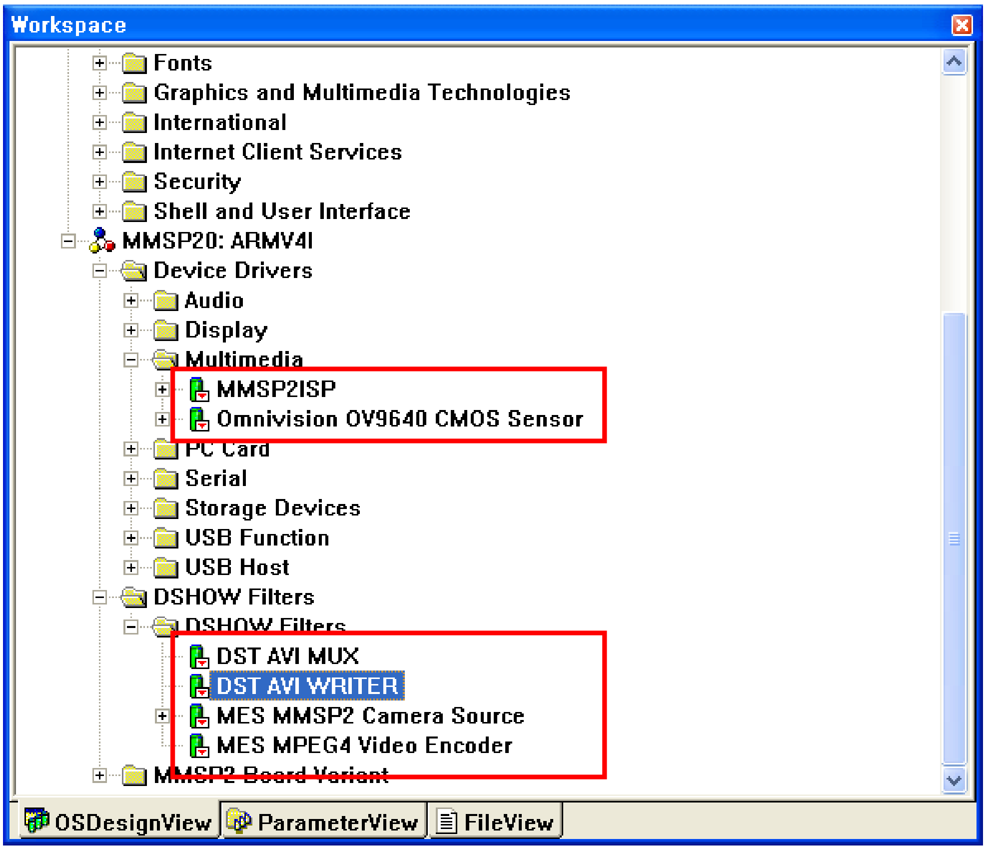
With this setting, NAND Flash Storage is mounted as root (“\”). And registry is stored in **\Documents and Settings** folder.

12. Camera Driver Guide

12.1 Adding Camera Driver

A. Add **Third Party** → **BSPs** → **MMSP20** → **Device Drivers** → **Multimedia** → **VidoeIn** → **MMSP2ISP**

B. Add **Third Party** → **BSPs** → **MMSP20** → **Device Drivers** → **Multimedia** → **VidoeIn** → **Omnivision OV9640 CMOS Sensor**



12.2 Adding Related Features

A. Add **Third Party** → **BSPs** → **MMSP20** → **DSHOW Filters** → **DSHOW Filters** → **MES MMSP2 Camera Source** component.

B. Add **Third Party** → **BSPs** → **MMSP20** → **DSHOW Filters** → **DSHOW Filters** → **MES MPEG4 Video Encoder** component.

C. Add **Third Party** → **BSPs** → **MMSP20** → **DSHOW Filters** → **DSHOW Filters** → **DST AVI MUX** component.

D. Add **Third Party** → **BSPs** → **MMSP20** → **DSHOW Filters** → **DSHOW Filters** → **DST AVI WRITER** component.

Add DirectShow related features in Core OS → Multimedia Technologies. See **MPEG4HW Encoder Filter Guide** section for more information.

12.3 Running Camera Test Application

When Camera component is included, a camera test application is also included in OS image. You can run this application on Windows CE Explorer. Open Explorer and change folder to \Windows, then run '**Camera**'.

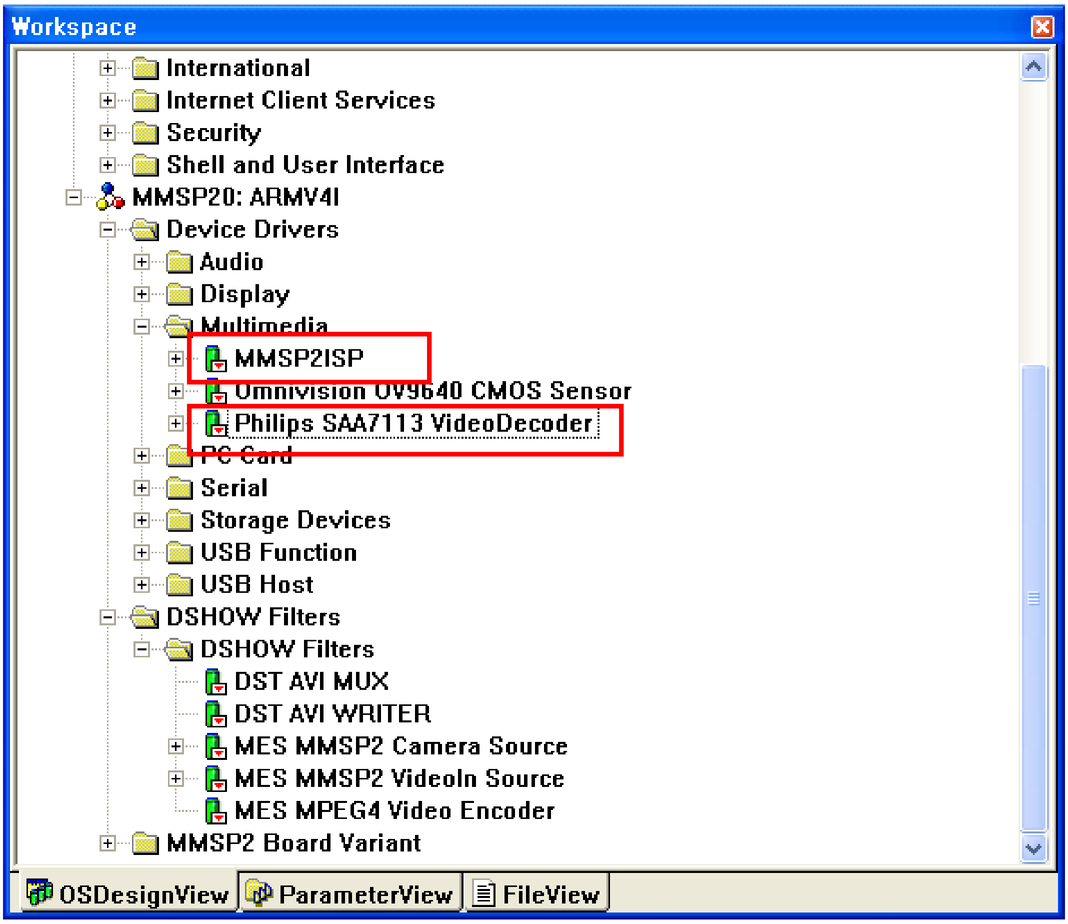
13. Video Decoder Driver Guide

13.1 Adding Video Decoder Driver

A. Add **Third Party** → **BSPs** → **MMSP20** → **Device Drivers** → **Multimedia** → **VidoeIn** → **MMSP2ISP**

B. Add **Third Party** → **BSPs** → **MMSP20** → **Device Drivers** → **Multimedia** → **VidoeIn** → **Philips SAA7113 VideoDecoder or Ti TVP5150A VideoDecoder.**

※ For DTK revision 4th or later, select Philips SAA7113 VideoDecoder.



13.2 Adding Related Features

A. Add **Third Party** → **BSPs** → **MMSP20** → **DSHOW Filters** → **DSHOW Filters** → **MES MMSP2 VideoIn Source** component.

B. Add **Third Party** → **BSPs** → **MMSP20** → **DSHOW Filters** → **DSHOW Filters** → **MES MPEG4 Video Encoder** component.

C. Add **Third Party** → **BSPs** → **MMSP20** → **DSHOW Filters** → **DSHOW Filters** → **DST AVI MUX** component.

D. Add **Third Party** → **BSPs** → **MMSP20** → **DSHOW Filters** → **DSHOW Filters** → **DST AVI WRITER** component.

Add DirectShow related features in Core OS → Multimedia Technologies. See **MPEG4HW Encoder Filter Guide** section for more information.

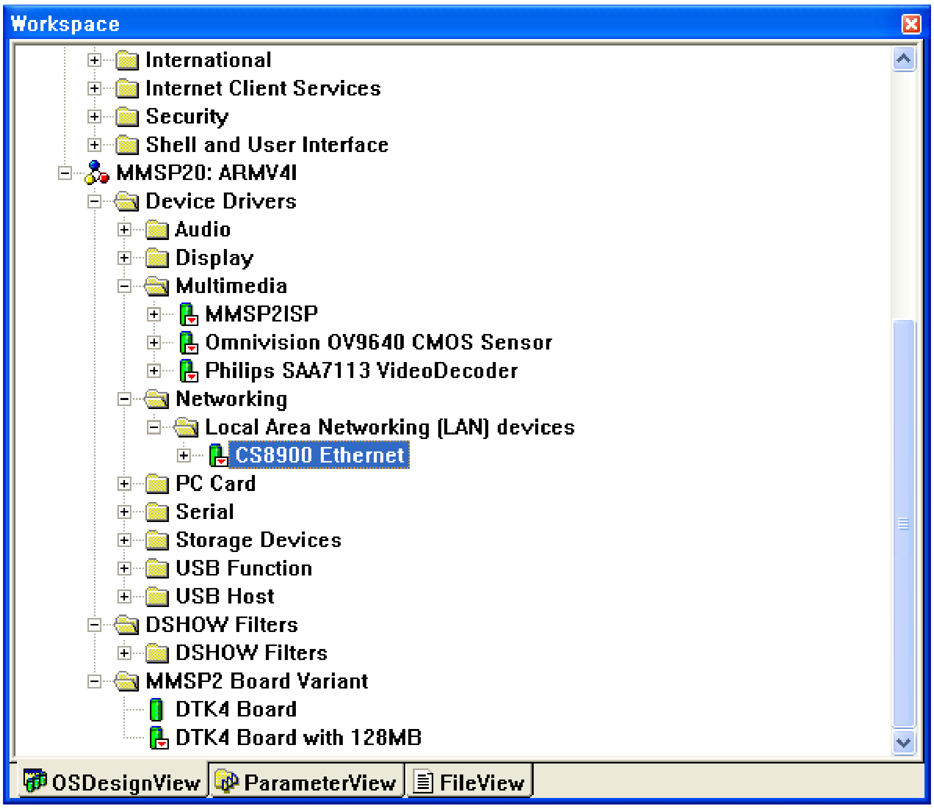
13.3 Running VideoIn Test Application

When Video decoder component is included, a VideoIn test application is also included in OS image. You can run this application on Windows CE Explorer. Open Explorer and change folder to \Windows, then run '**VideoIn**'.

14. Local Area Networking Controller Driver (CS8900) Guide

14.1 Adding CS8900 Driver

Add **Third Party** → **BSPs** → **MMSP20** → **Device Drivers** → **Networking** → **Local Area Networking** → **CS8900 Ethernet** component.



14.2 Available Features with Ethernet Driver

- Connect to Web by using Internet Explorer.
- Files in DTK board can be accessed when FTP Server component is included.

14.3 Setting IP Address

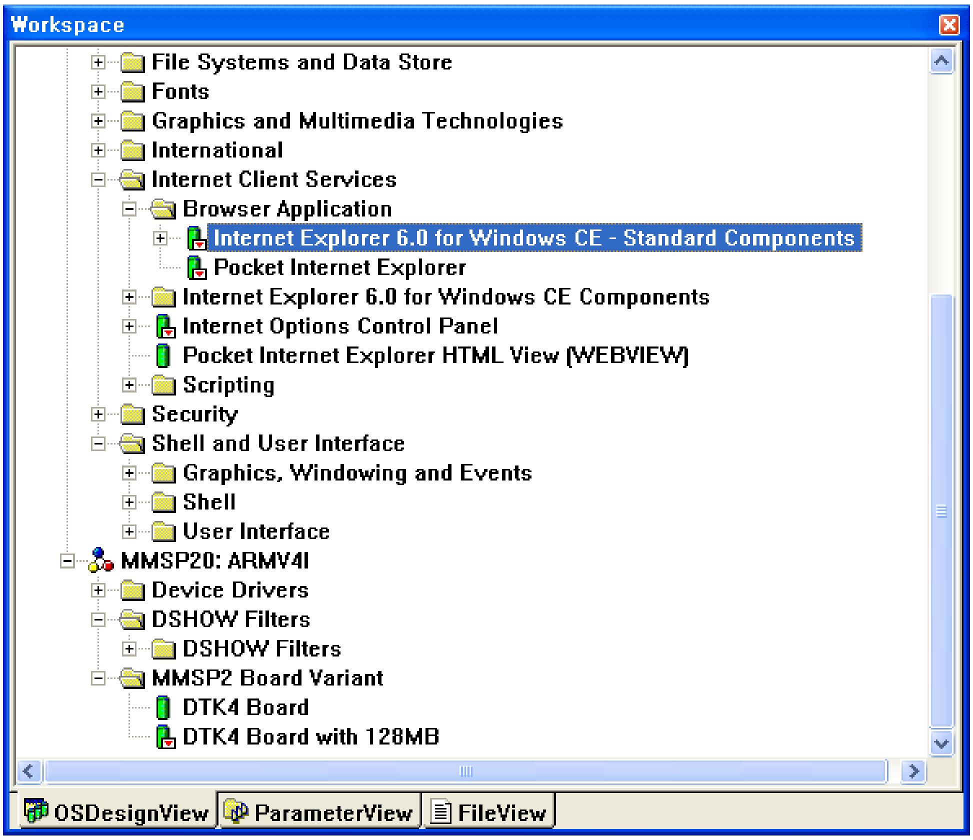
To set IP address, open **ends4isa.reg** in **WINCE500\PLATFORM\MMSP20\DRIVERS\NETCARD\CS8900** folder and modify value marked as dark color.

```
[HKEY_LOCAL_MACHINE\Comm\ENDS4ISA1\Parms\TcpIp]
   ; This should be MULTI_SZ
   "DefaultGateway"="220.78.49.252"
```

```
; This should be SZ... If null it means use LAN, else WAN and Interface.
"LLInterface"=""
; Use zero for broadcast address? (or 255.255.255.255)
"UseZeroBroadcast"=dword:0
; Thus should be MULTI_SZ, the IP address list
"IpAddress"="220.78.49.136"
; This should be MULTI_SZ, the subnet masks for the above IP addresses
"Subnetmask"="255.255.255.128"
"DNS"="168.126.63.1"
"EnableDHCP"=dword:0
```

14.4 Networking Related Components

- Select one of IE 6.0 or Pocket IE in **Internet Client Services** → **Browser Application** to use Internet Explorer.



- Add **FTP Server** component in **Communication Services and Networking** → **Servers** of the catalog. Add registry entry shown below to the **platform.reg** or **project.reg** to enable anonymous login to FTP server. (You can edit these files by selecting them on the **ParameterView** tab of the Platform Builder window.)

```
; @CESYSGEN IF SERVERS_MODULES_FTPD
; @CESYSGEN IF SERVERS_MODULES_SERVICES
[HKEY_LOCAL_MACHINE\Services\FTPD]
; @CESYSGEN ELSE
; @CESYSGEN ENDIF SERVERS_MODULES_SERVICES
  "Dll"="FTPD.Dll"
  "Order"=dword:9
  "Keep"=dword:1
  "Prefix"="FTP"
  "Index"=dword:0
```

```
[HKEY_LOCAL_MACHINE\COMM\FTPD]
   "IsEnabled"=dword:1
   "UseAuthentication"=dword:1
;   "UserList"="add;semicolon;separated;list;of;users;here"
   "AllowAnonymous"=dword:1
   "AllowAnonymousUpload"=dword:1
   "AllowAnonymousVroots"=dword:1
   "DefaultDir"="\\"
;To control logging
   "DebugOutputChannels"=dword:2
   "DebugOutputMask"=dword:17
   "BaseDir"="\\Windows"
   "LogSize"=dword:1000
; @CESYSGEN ENDIF SERVERS_MODULES_FTPD
```

15. Button Device Driver Guide

Button device driver gets key input and let a application know what key is pressed. The source code is located at **MMSP20/DRIVERS/REMOTECTLR**. And we provide a simple application to test key input (**MMSP20/APPS/ButtonTest**)

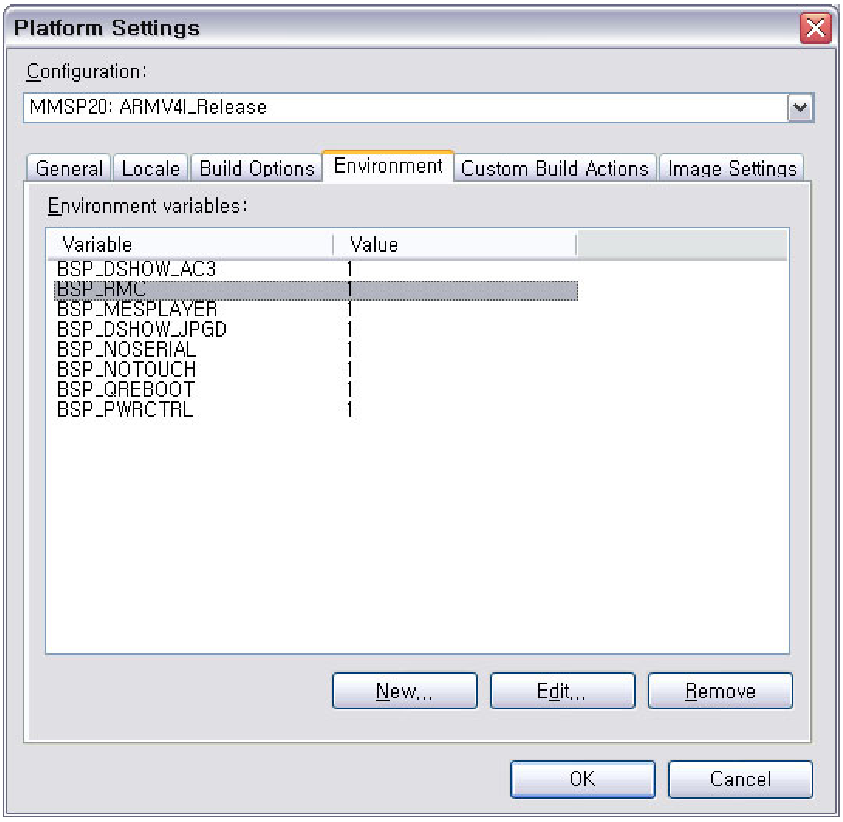
DTK4 development kit that has 5 general purpose keys (SW3 ~ SW7).

Key Assignments	
SW3	UP (VK_UP)
SW4	DOWN (VK_DOWN)
SW5	MENU (VK_MENU_
SW6	EXIT (VK_ESCAPE)
SW7	ENTER (VK_RETURN)

When you press a key it generates a KEY event.

15.1 Adding Button Device Driver

To add button device driver, you have to set **BSP_RMC** variable as 1.



15.2 Known problem(s)

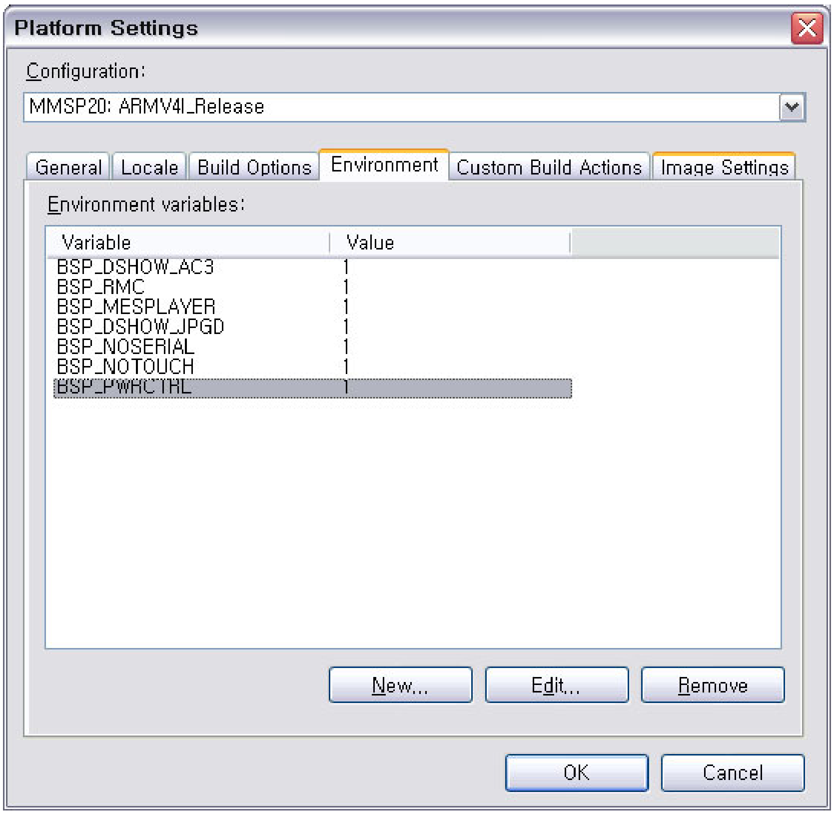
VK_MENU event is not passed to application.

16. Power Manager Driver Guide

16.1 Adding Power Manager Driver

Select **Platform -> Settings** menu and open **Environment** tab.

Add **BSP_PWRCTRL** variable and set value as **1**.



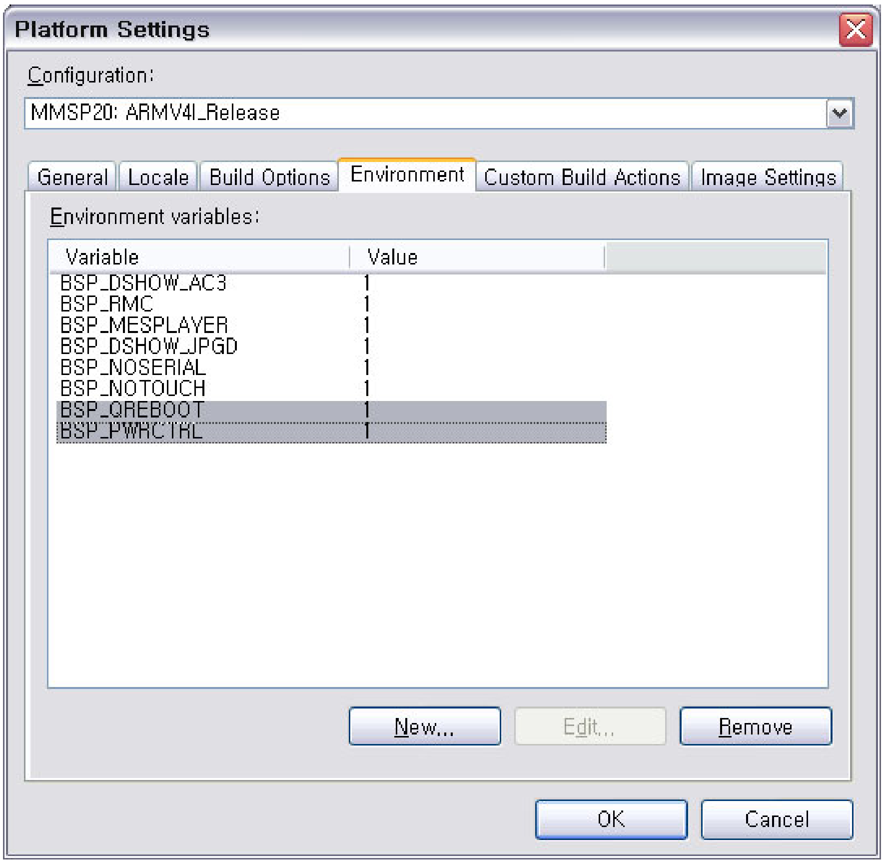
After setting environment variable, build OS image to apply it.

16.2 Using Suspend/Wakeup Function

After setting environment variable and building OS is done, you can use **EXTIN0** and **EXTIN1** button of the DTK 4.11 board for suspend/wakeup. To suspend system, press **EXTIN0** button. To wakeup suspended system, press **EXTIN1** button.

16.3 Quick Booting Mode

You can use quick booting mode *instead of* normal suspend/wakeup mode. Add both **BSP_PWRCTRL** and **BSP_QREBOOT** environment variable and set value of these variables as **1**.



After setting environment variable, build OS image to apply it.

16.4 Using Quick Booting Mode

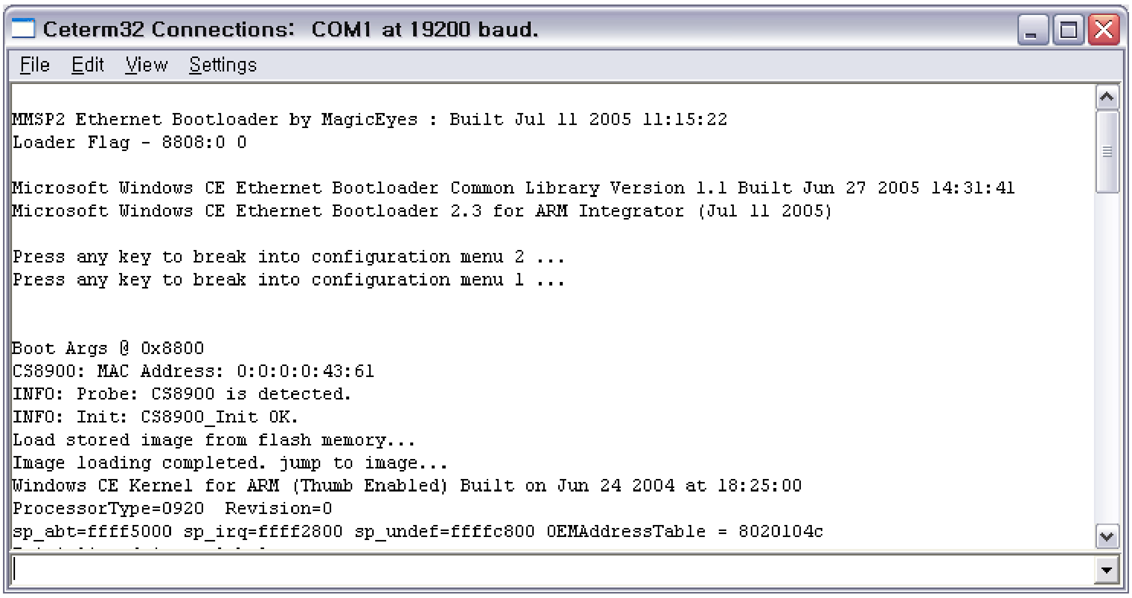
After setting environment variable and building OS is done, you can use **EXTIN0** and **EXTIN1** button of the DTK 4.11 board for quick reboot. To suspend system, press **EXTIN0** button. And when you press **EXTIN1** button at suspend state, system enters into quick boot state.

Quick rebooting is done with this sequence:

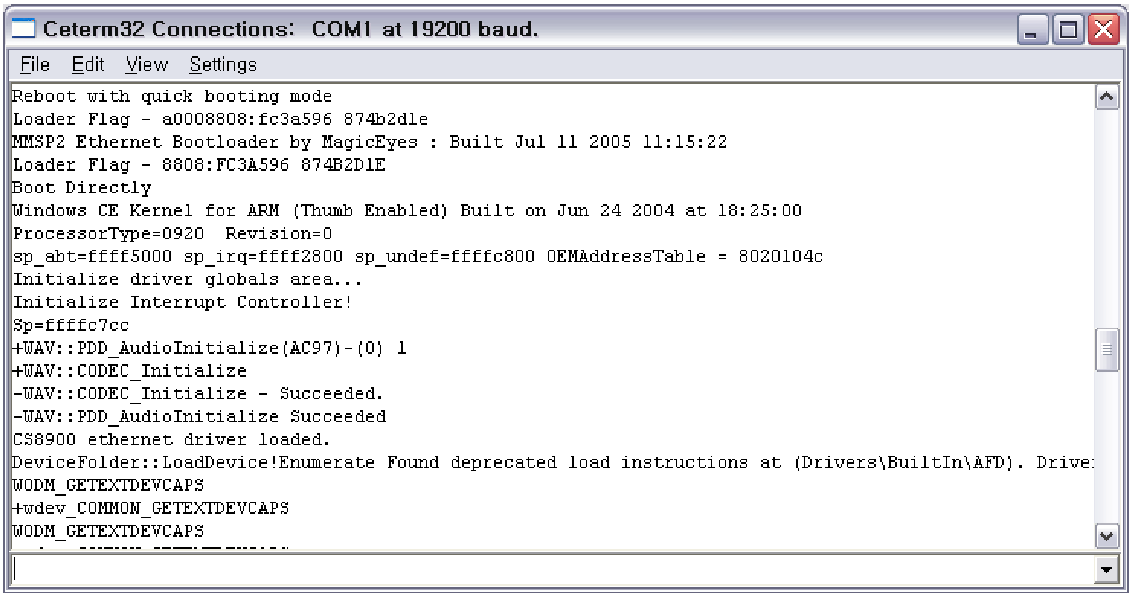
- When EXTIN1 button is pressed at suspend state, power manager driver sets quick reboot flag and reset system.
- Ethernet bootloader reads flag.
- If quick reboot flag is set, Ethernet bootloader jumps to OS base address without loading process.

Quick reboot time can be less than 5 seconds if OS does not have drivers or components that take long time for initialization.

This is an example of normal boot. (Boot from image stored in NAND flash)



This is an example of quick reboot. (Boot without OS image loading. See loader flag.)



17. MPEG4HW Decoder Guide

17.1 Specifications

- Supports AVI format only
- Supports MPEG4 Simple Profile and Advanced Simple Profile
- Supports DivX 4.x, DivX 5.x, DivX 3.11 and Xvid

17.2 Performance

- Please see MPEG4 Decoder performance chart.

17.3 Limitations

- Do not support 4MV + QPEL contents.
- Do not support GMC with 2 or more warping points.
- In some DivX 3.11 contents you may see defections.

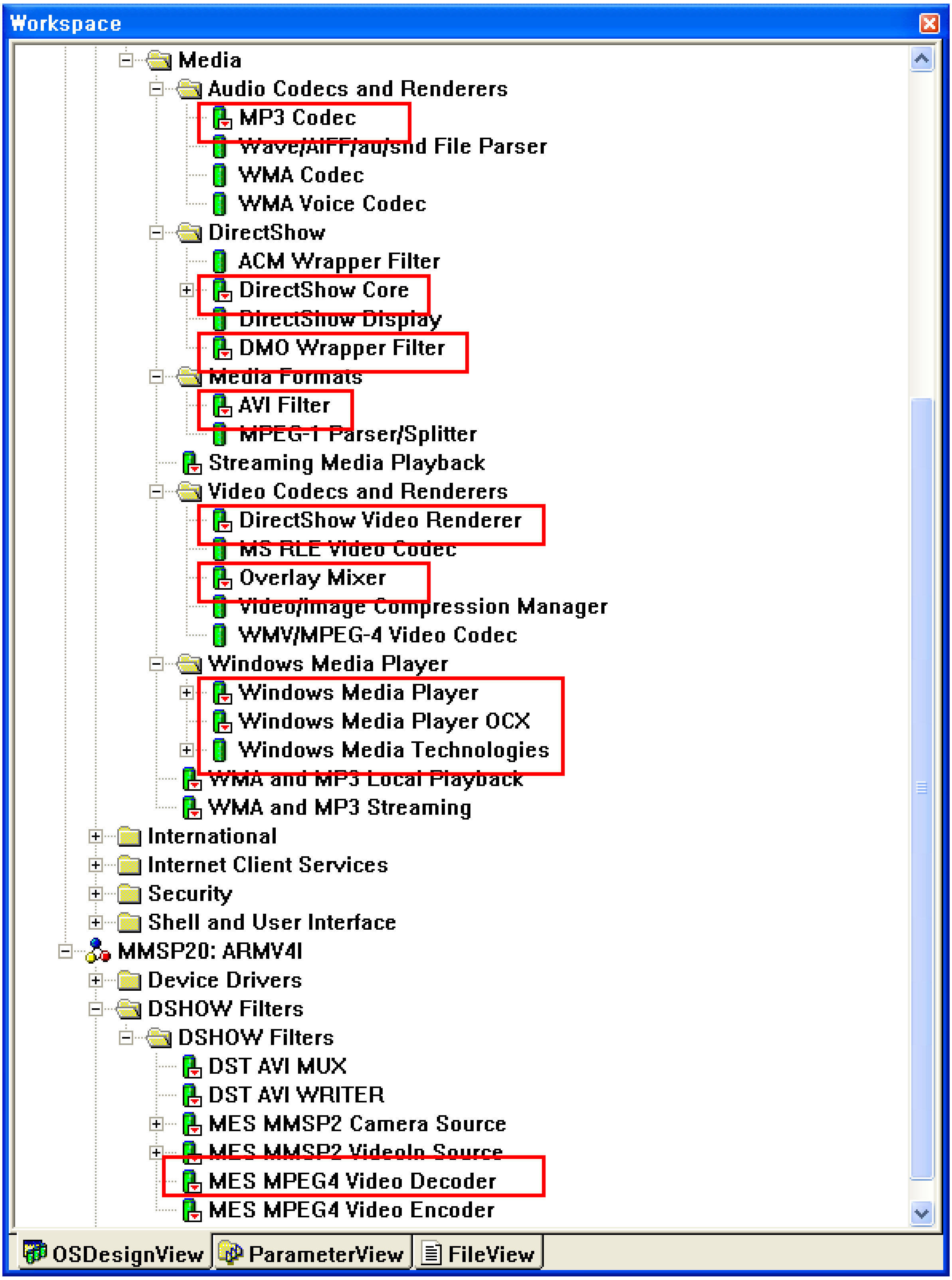
17.4 Required Components

See Display Driver Guide to see information about output of decoded result.

Add components shown below in **Core OS** → **Windows CE Devices** →**Graphics and Multimedia Technologies** → **Media** of the catalog.

- **DirectShow** → **DirectShow Core**
- **DirectShow** → **DMO Wrapper Filter**
- **Audio Codecs and Renderers** → **MP3 Codec**
- **Media Formats** → **AVI Filter**
- **Video Codecs and Rednerers** → **Overlay Mixer**
- **Video Codecs and Rednerers** → **DirectShow Video Renderer**
- **Windows Media Player** → **Windows Media Player**
- **Windows Media Player** → **Windows Media Player OCX**
- **Windows Media Player** → **Windows Media Technologies**

Add **MMSP20: ARMV4I** → **DSHOW Filters** →**DSHOW Filters** →**MMSP2 MPEG4 Video Decoder**.


Workspace
Media
Audio Codecs and Renderers
MP3 Codec
Wave/AIFF/au/snd File Parser
WMA Codec
WMA Voice Codec
DirectShow
ACM Wrapper Filter
DirectShow Core
DirectShow Display
DMO Wrapper Filter
Media Formats
AVI Filter
MPEG-1 Parser/Splitter
Streaming Media Playback
Video Codecs and Renderers
DirectShow Video Renderer
MS RLE Video Codec
Overlay Mixer
Video/Image Compression Manager
WMV/MPEG-4 Video Codec
Windows Media Player
Windows Media Player
Windows Media Player OCX
Windows Media Technologies
WMA and MP3 Local Playback
WMA and MP3 Streaming
International
Internet Client Services
Security
Shell and User Interface
MMSP20: ARMV4I
Device Drivers
DSHOW Filters
DSHOW Filters
DST AVI MUX
DST AVI WRITER
MES MMSP2 Camera Source
MES MMSP2 VideoIn Source
MES MPEG4 Video Decoder
MES MPEG4 Video Encoder
OSDesignView
ParameterView
FileView

18. DST WMV Decoder

18.1 Specifications

- Supports QVGA, 500Kbps
- Supports WMV7, WMV8, WMV9

For more information see DSTWMVDec.doc in MMSP20\DOCS folder.

18.2 Limitations

18.3 Required Components

- All components for MPEG4HW Decoder in **Core OS → Windows CE Devices →Graphics and Multimedia Technologies → Media**
- **Core OS → Windows CE Devices →Graphics and Multimedia Technologies → Media → Streaming Media Playback**
- **Third Party →BSPs →MMSP20:ARMV4I →DSHOW Filters → DSHOW Filters →DST WMV DECODER**

19. MES MP3 Decoder

MES MP3 Decoder evaluation version is included. It has 300seconds time-limit.

It supports VBR MP3 in AVI file and gives better performance in video decoding than Microsoft MP3 decoder.

19.1 Required Components

- **Third Party →BSPs →MMSP20:ARMV4I →DSHOW Filters → DSHOW Filters →MES MP3 Decoder Eval.**

20. Graph Editer

Windows CE 5.0 Platform Builder doesn't support __SYSGEN_GRAPHEDT environment setting value.

20.1 How to include

PBWorkspaces\"%YOUR PROJECT"\wince500\ MMSP20_ARMV4I\OAK\MISC\cesysgen.bat

modify value marked as dark color

```
@echo off
call %_PUBLICROOT%\cebase\oak\misc\cesysgen.bat %*
set DIRECTX_MODULES=%DIRECTX_MODULES% graphedt
echo %DIRECTX_MODULES%
goto :EOF
```

21. MESPlayer

MESPlayer is a simple media player developed by MagicEyes.Co.Ltd. for application developer. It can play a MPEG4 AVI file and encode camera input. Source code also is included in BSP.

Location of source file is MMSP20\APPS\MESPlayer.

21.1 How to Build

To build MESPlayer, see a document in MMSP20\APPS\MESPlayer folder.

21.2 How to Add MESPlayer into Windows CE image

Copy your new MES.exe into MMSP20\FILES folder.

Copy and paste following lines into project.bib. (You can edit this file by selecting it on the **ParameterView** tab of the Platform Builder window.)

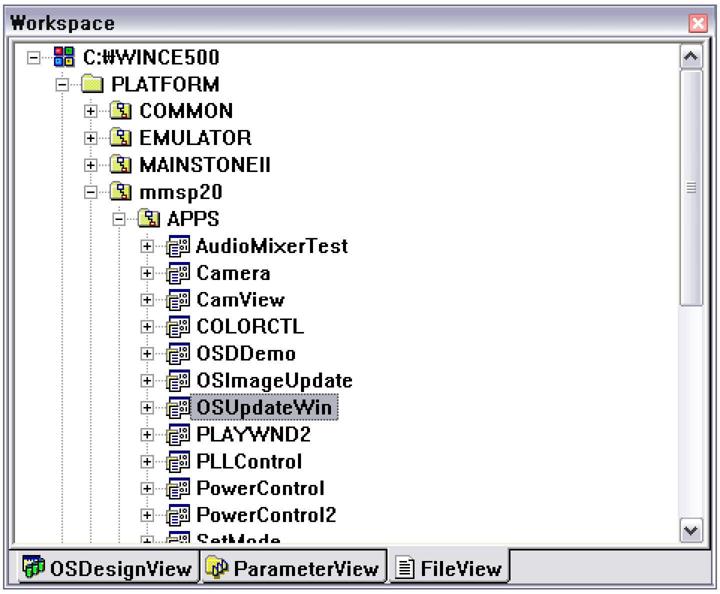
```
FILES
; Name          Path                                        Memory Type
; -------------  -------------------------------------------  -----------
Mes.exe        $(_WINCEROOT)\PLATFORM\MMSP20\FILES\Mes.exe      NK
```

22. OS Update Application

OS update application (OSUpdateWin) is a sample program that replaces OS image in NAND flash of the MMSP2 based system. This application works with Ethernet bootloader of MMSP2 BSP.

22.1 How to Build

Source files of this application are in **APPS\OSUpdateWin**. Click this folder in FileView of the Windows CE Platform Builder window (with right button of the mouse) and select **Build Current Project**.



22.2 How to Add OSUpdateWin into Windows CE Image

You can add OSUpdateWin.exe to Windows CE OS image by adding environment **BSP_OSUPDATE** and set value as **1**. When OS is built with this setting, Shortcut of this application is shown on the desktop.

22.3 Using OSUpdateWin Application

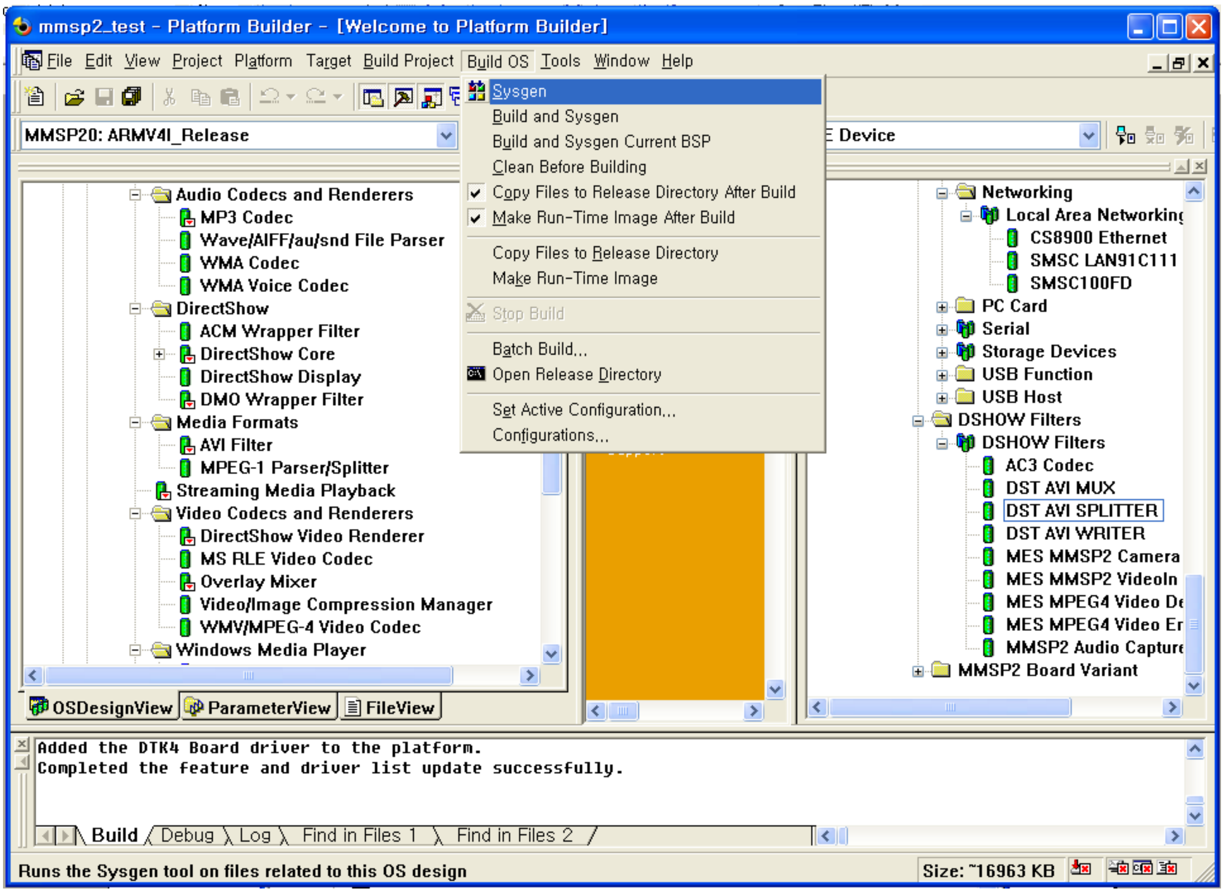
When this application stars it shows file selection window. You can select a valid Windows CE OS image file. This application supports ROM file format. Basically, this format of OS image file is generated as “NK.NB0” during Sysgen or Makeimage phase.

After file selection is done, this application updates OS image in NAND flash of the system. When it completes it shows a pop-up message of completion.

23. Building WinCE OS

23.1 Building Entire OS

When setting is completed, start building by selecting **Build OS** -> **Sysgen**.



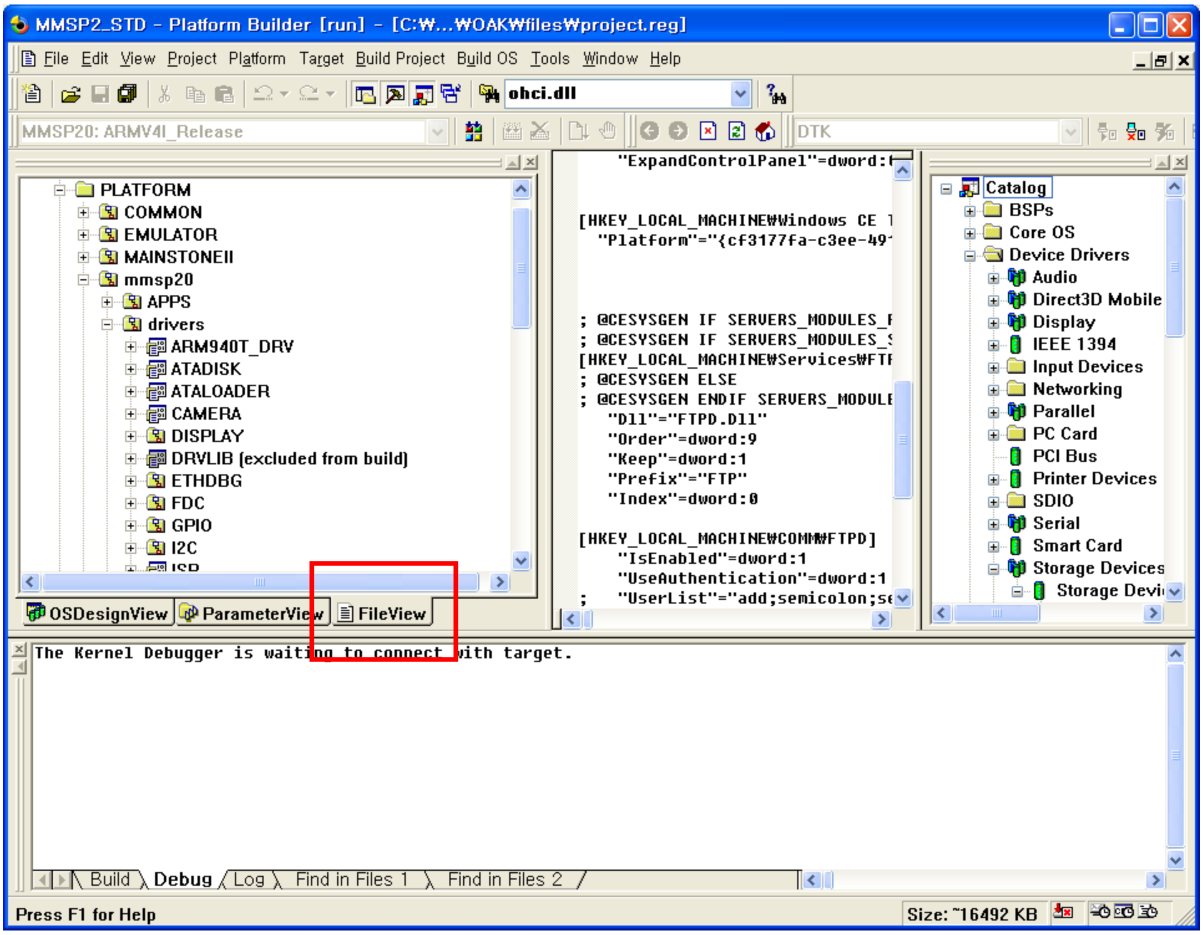
If build is done without error, download new OS image to DTK board. (See next chapter.)

23.2 Building Part of BSP Source

When you develop your own Windows CE OS image, you can build part of BSP source which you modified. By this partial building you can reduce build time dramatically.

Caution: This build method should be applied only after building entire OS is done..

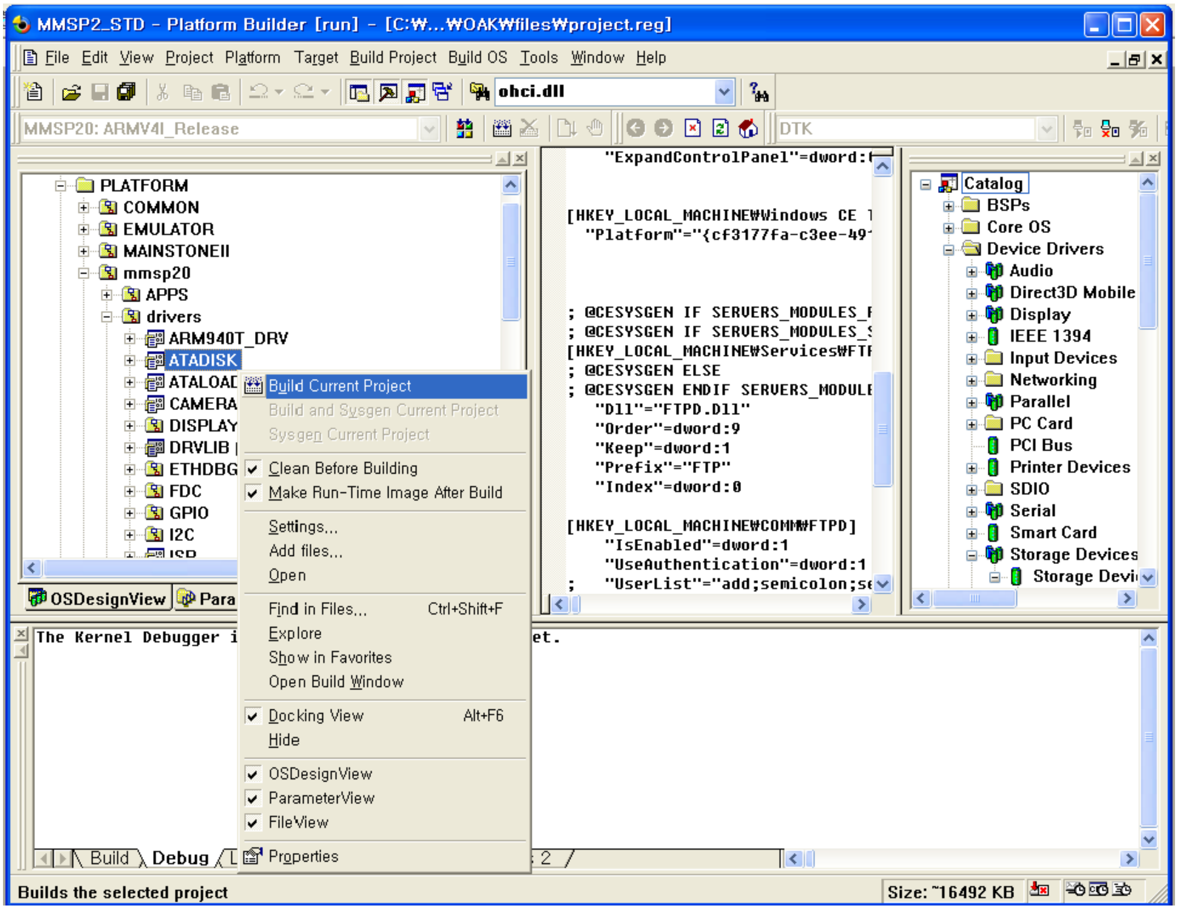
Select **File View** tab.



Select a folder that you want to build and press right button of the mouse.

When context menu is shown, select **Build Current Project**.

If **Make Run-time Image After Build** option is selected, OS image is generated after selected part of source is built and applied.



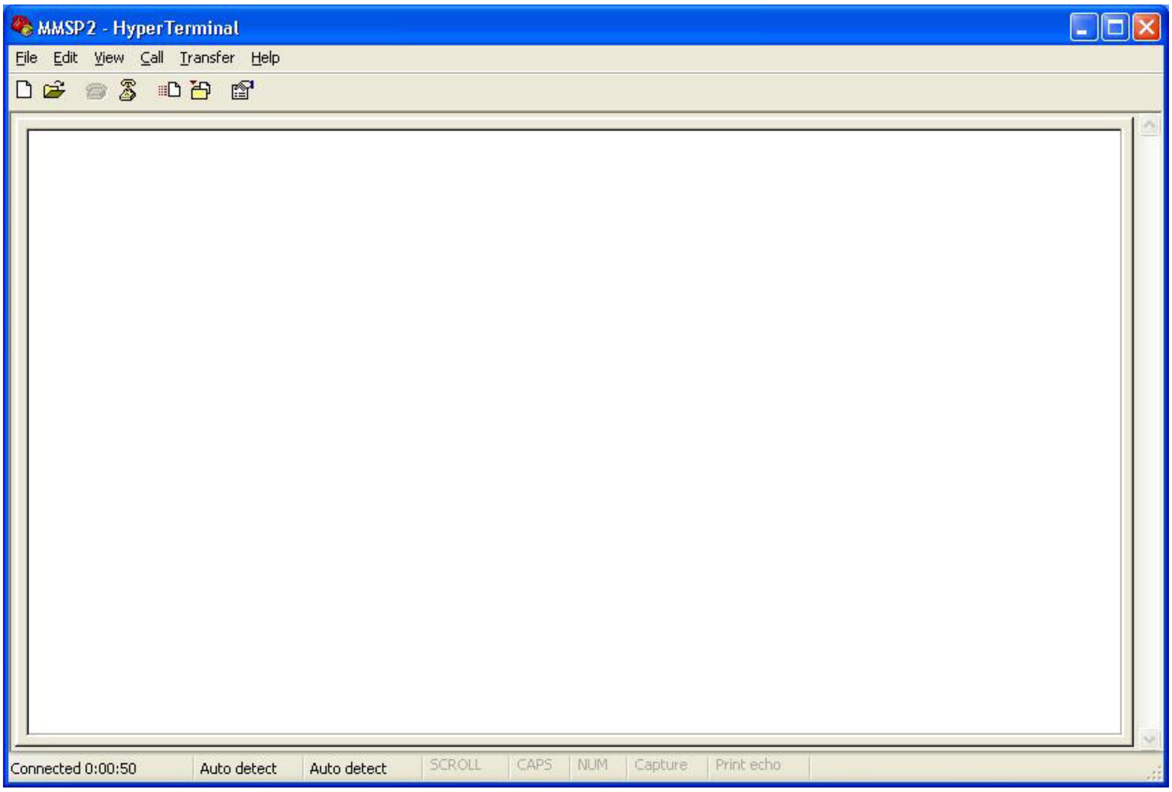
Above picture shows the process of building ATA Hard Disk driver. After building is done without error, new driver DLL file is automatically copied to release directory and making OS image is performed.

24. Download Image with Ethernet Bootloader

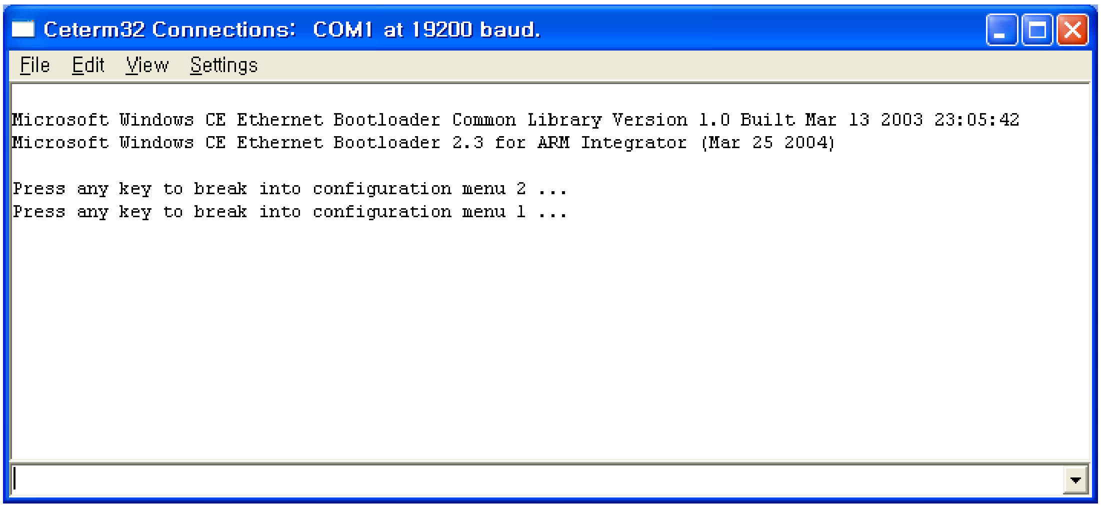
To download Windows CE OS image, make sure ethernet bootloader is correctly installed on DTK board.

24.1 Setting Configuration of Ethernet Bootloader

- Connect PC serial port and debugging serial port of the DTK board with cross cable.
  (Debugging port is right of the two serial ports.)
- Connect LAN Cable on the board.
- Run terminal program on PC and set baud rate to 19200bps.
  EX) START – Programs – Accessories – Communications – HyperTerminal



- When connection and setting is done correctly and power of the DTK board is on, bootloader message is displayed on terminal screen. (Hyper Teminal or Ceterm)



24.2 Settings for OS Image Download

Go to menu by pressing any key when “Press any key to break into configuration menu” message is shown.

24.2.1 Setting Boot Options

Boot options are:

- **Mode1.** Boot with OS image stored in NAND flash memory.
- **Mode2.** Download OS image through Ethernet and store in NAND flash. Boot after writing to flash memory is completed.
- Download OS image through Ethernet and boot without storing to NAND flash.

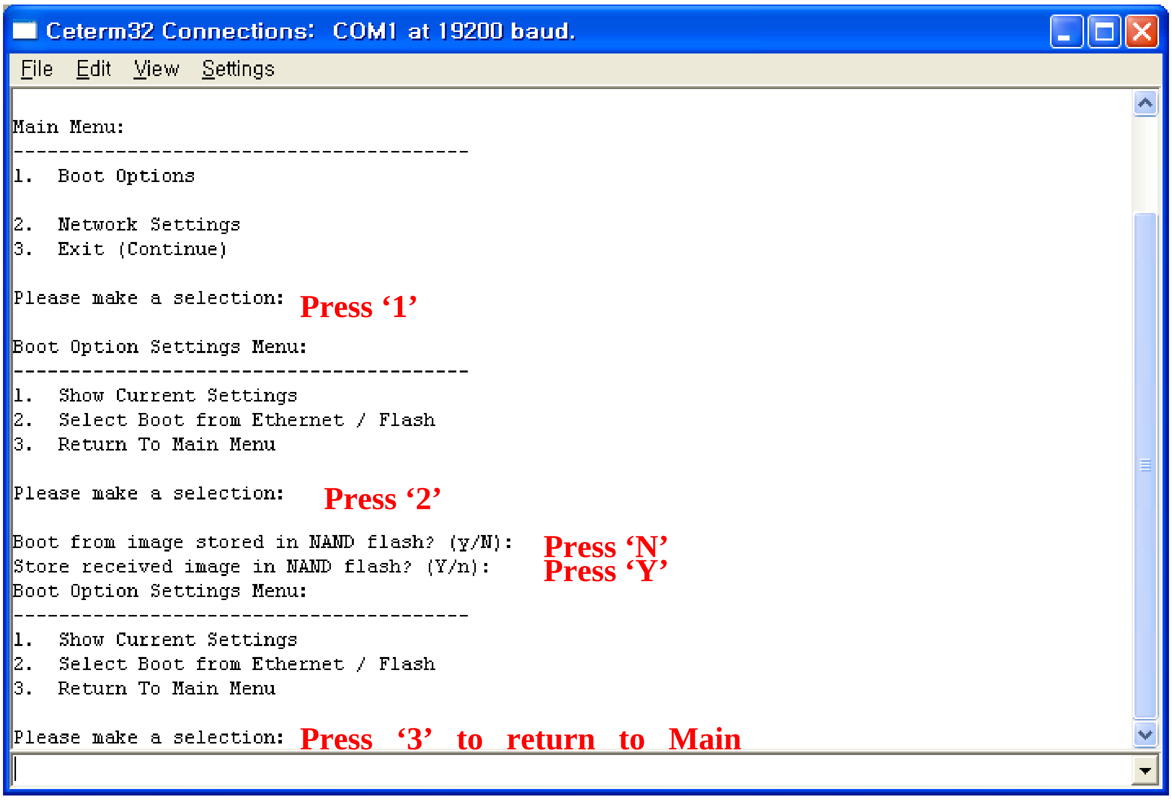


Figure. Mode2-Download OS Image mode

24.2.2 Setting Network Options

Set IP and MAC address for OS image download through Ethernet.

Select **Network Settings** -> **Store Static IP Address and Subnet Mask** and Set IP Address, Subnet mask, MAC address in turn.

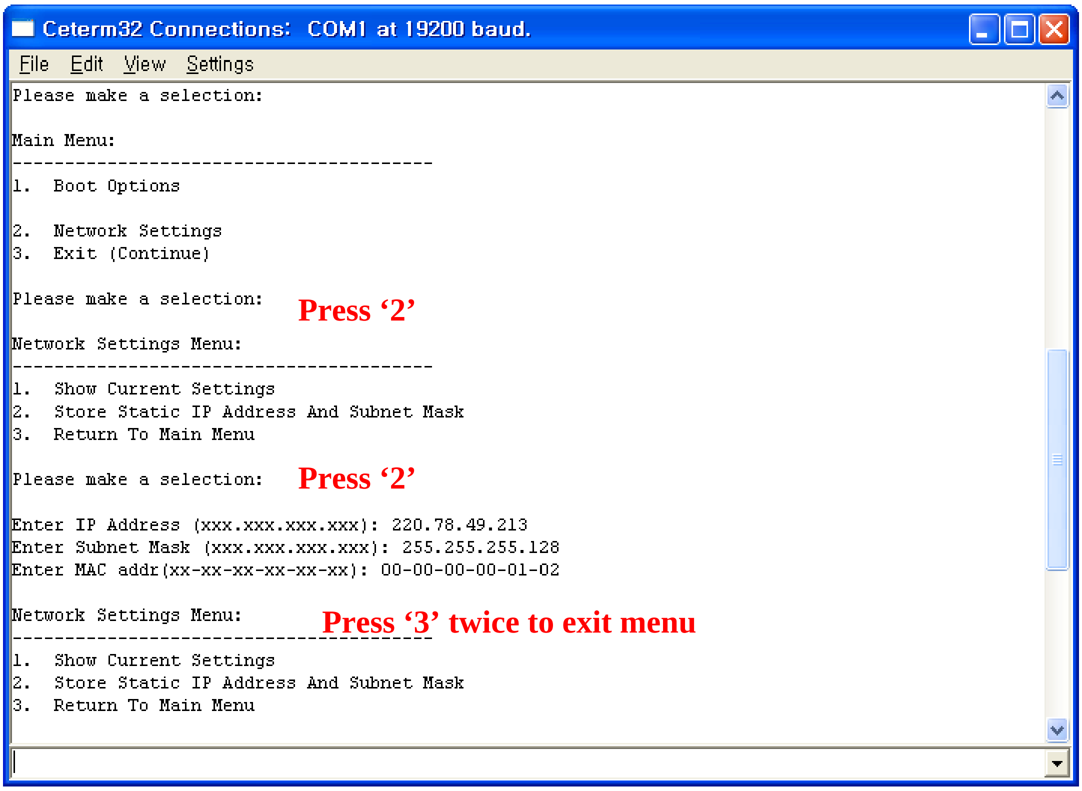


Figure. Network Setting

Finishing Configuration Setup

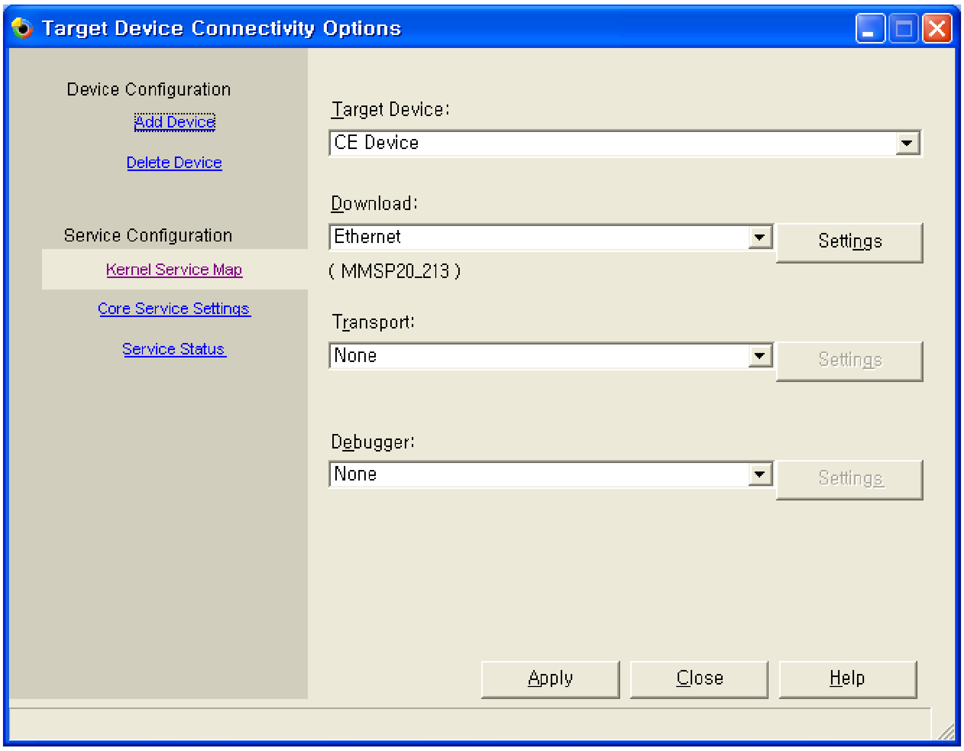
Save options by selecting **Return to Main Menu** -> **Exit**.

Now bootloader will be run according to the stored options.

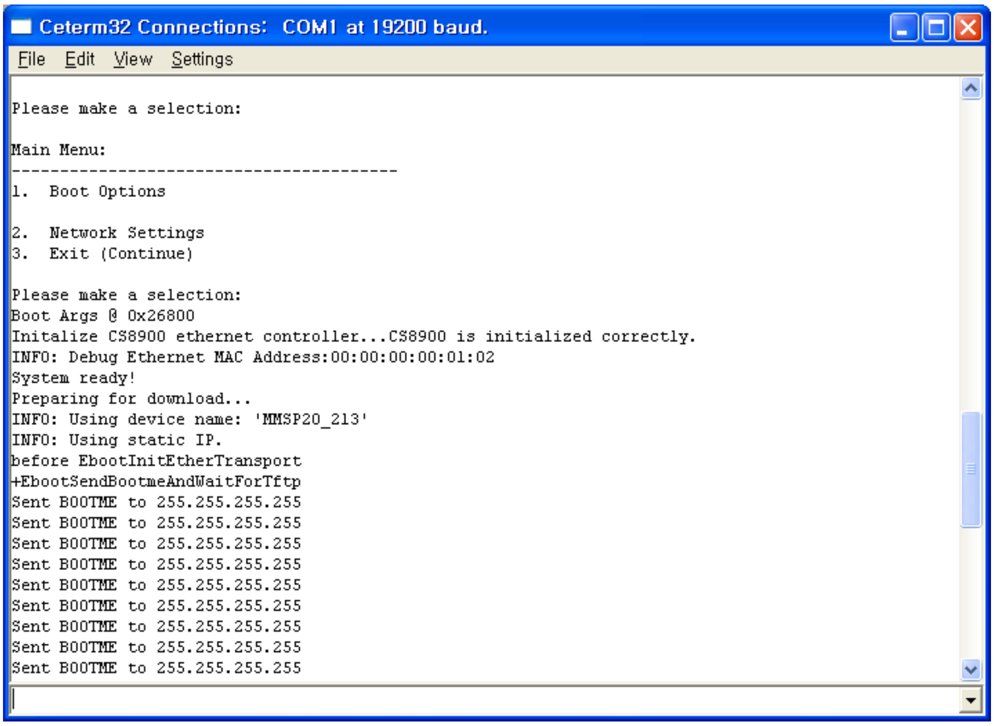
Setting Platform Builder for OS Image Download

Select **Target** -> **Connectivity Options** in menu.

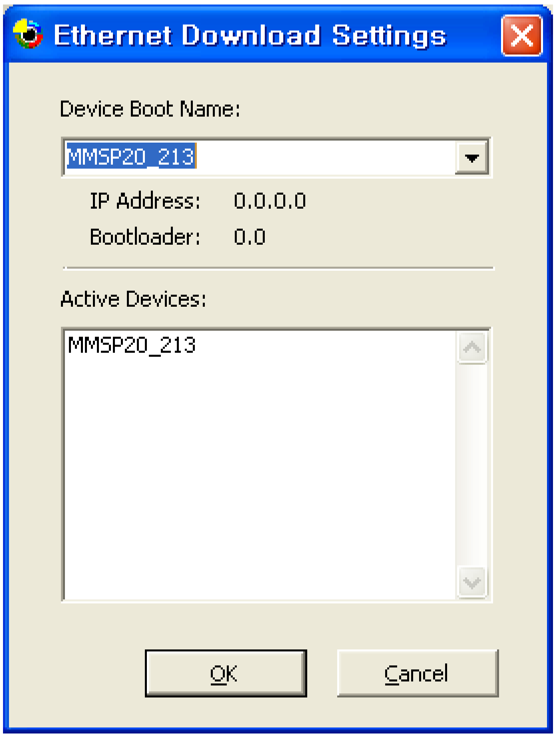
Set **Download** as ‘Ethernet’ and prepare to select download target by pressing ‘Settings’ button.



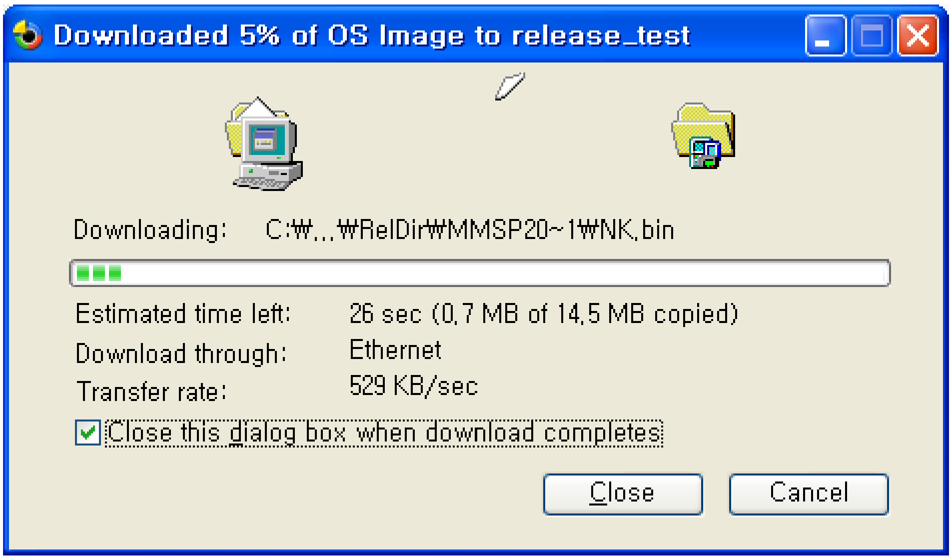
Bootloader sends BOOTME packet when boot mode is set as Ethernet download mode.



Device name of the target board is shown on the **Active Devices** list when Platform builder receives BOOTME packet from the board. Select one from the list.



Finish configuration and run Ethernet download by selecting **Target** -> **Attach Device**. (Starting download is possible only when bootloader sends BOOTME packet.)



After downloading is completed, OS image runs. If you selected option that store received image to flash memory, it takes some time to store image and run OS.

24.2.3 NAND Flash Mode Setting

After store OS Image, User must set Mode 1 to boot NAND Flash mode

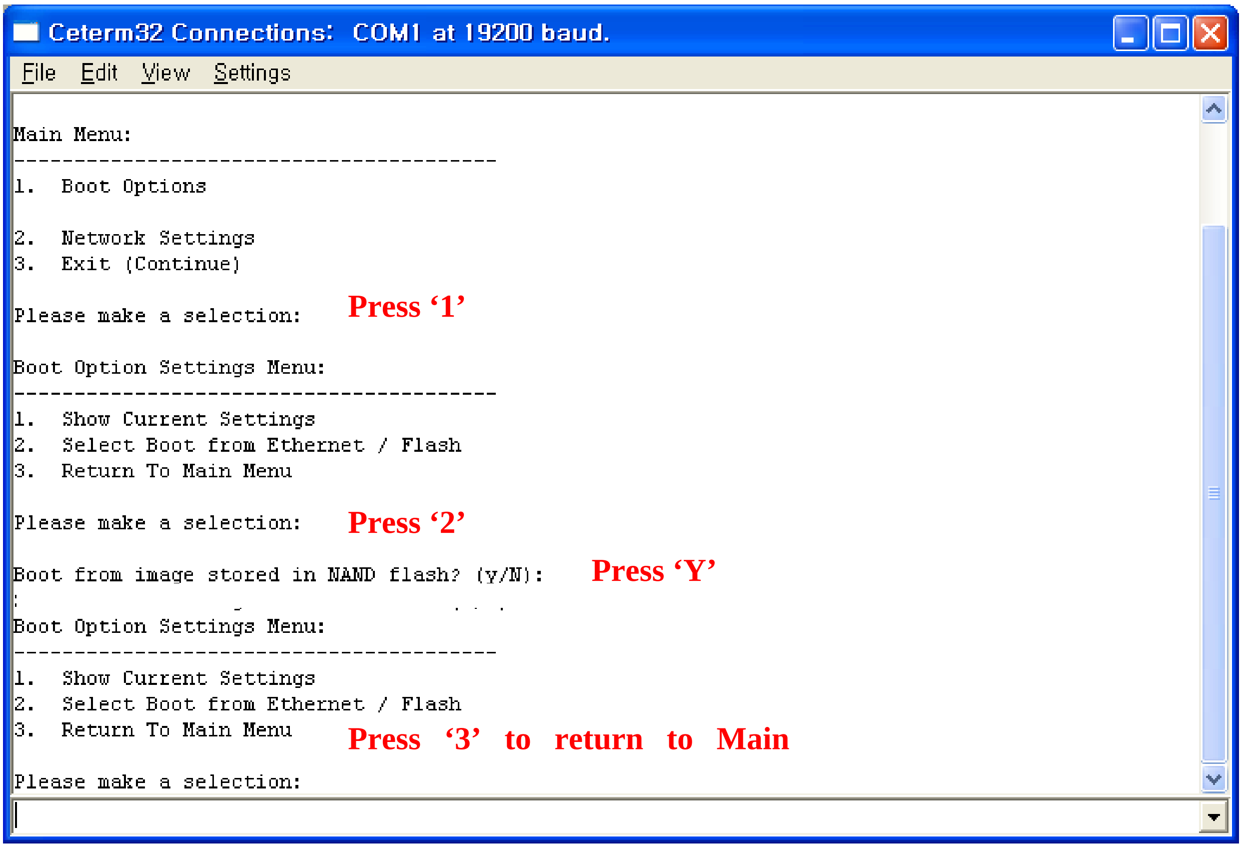
- 

Figure. Mode1-NAND Flash mode

25. Installing Bootloader Image to NAND Flash

Bootloader for Windows CE image is installed when DTK board is delivered. You can replace this bootloader image with ARM debugging tools and DTK diagnostic program.

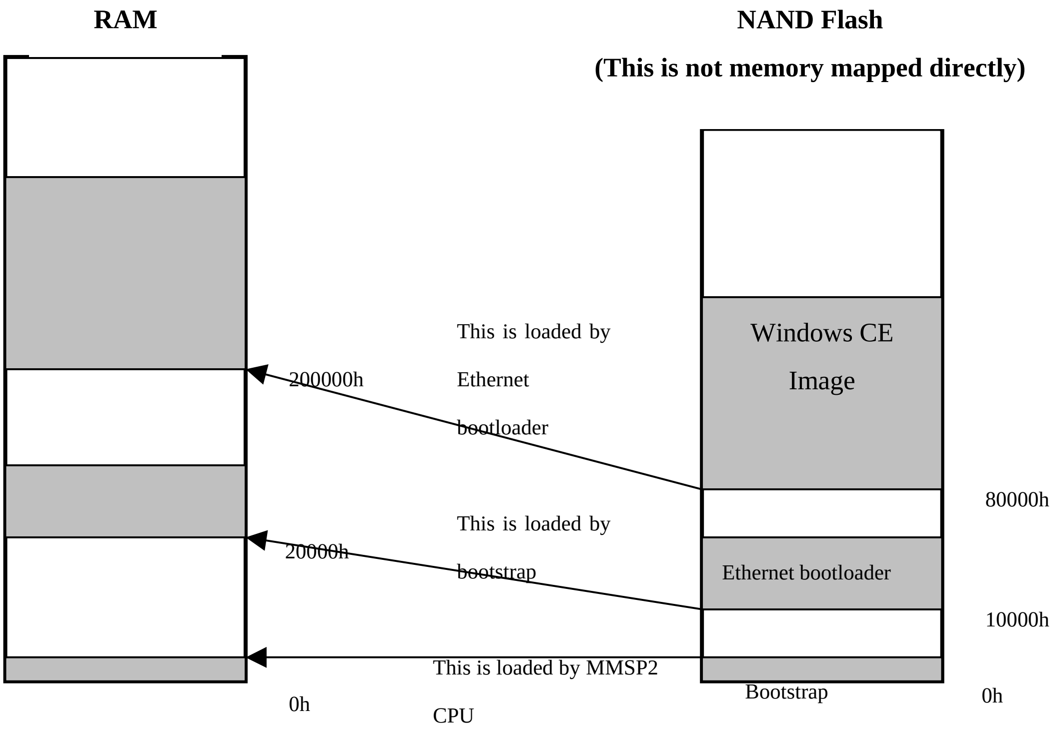
25.1 Boot Sequence

Current version of bootloader used on MMSP2 DTK board is divided with two steps of images. First bootlaoder is a small image (less than 512 bytes) and called ‘bootstrap.’ This bootstrap is loaded by MMSP2 CPU and it launches second bootloader. This second bootloader is Ethernet bootloader that you have seen in previous chapter.

E. When power is on, MMSP2 CPU loads bootstrap stored in address 0h of NAND flash. This bootstrap image is located in address 0h of the RAM.

F. As bootstrap runs, it loads Ethernet bootloader image stored in 10000h of NAND flash. This Ethernet bootloader image is located in address 20000h of the RAM.

G. Ethernet bootloader loads Windows CE image from NAND flash or get new image through Ethernet.



25.2 Installing Bootloader Image

Because address of NAND flash is not memory mapped directly, a program that accesses to NAND flash interface is needed to write images in NAND flash.

25.2.1 Requirements

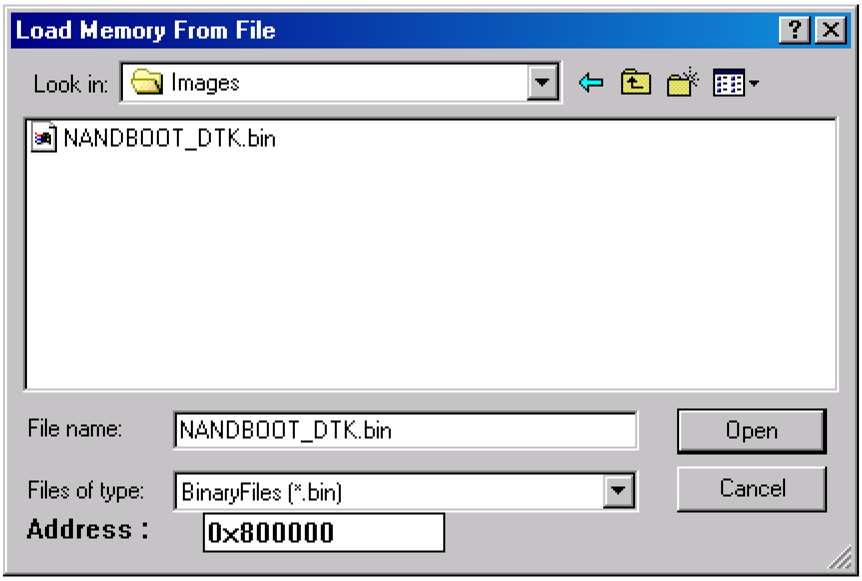
To install or replace bootloader image, you must have:

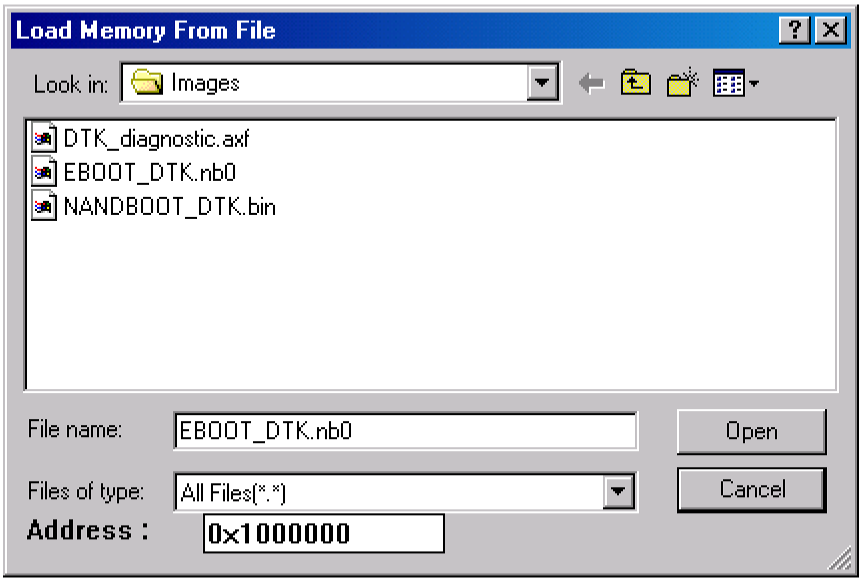
- A JTAG emulator (OPENice32-A900 or OPENice-A1000)
- A debugger (Spider)
- MMSP2 DTK diagnostic program image

25.2.2 Installing bootloader images

A. Run Spider.

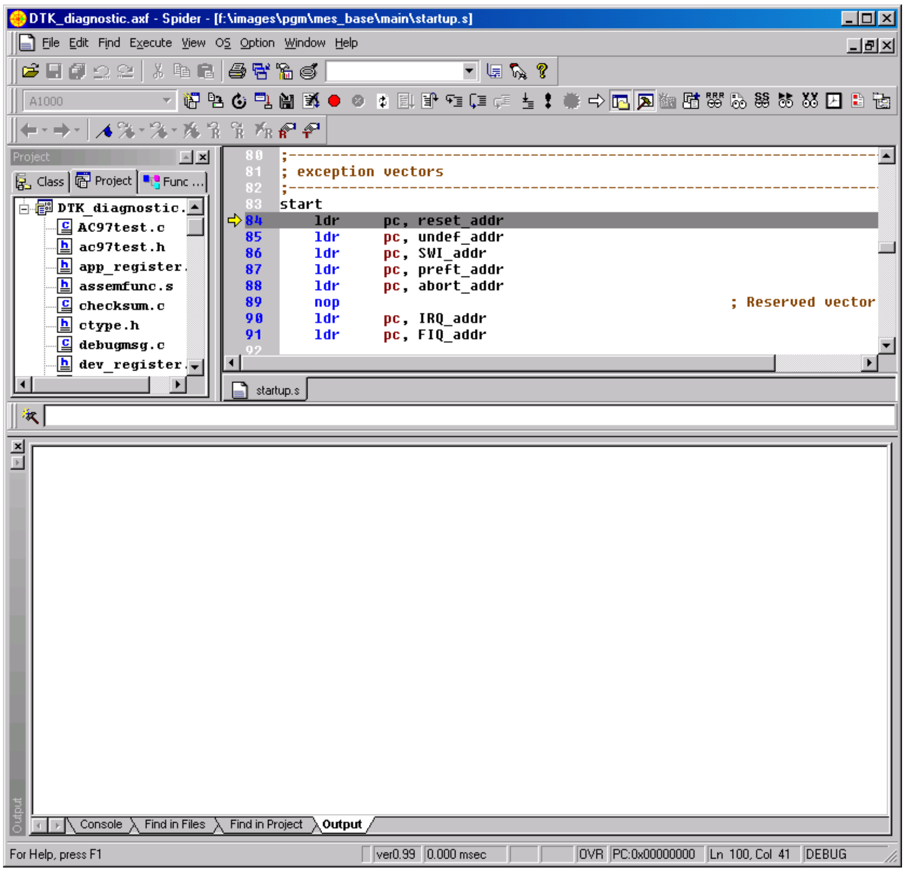
B. Select **File** -> **Load Memory From File** to load new bootloader image to RAM. Address can be any address of RAM area that does not have any interference with DTK diagnostic program. In this guide, we load NANDBOOT_DTK.bin (bootstrap image) to 800000h and EBOOT.nb0 (Ethernet bootloader image) to 1000000h.



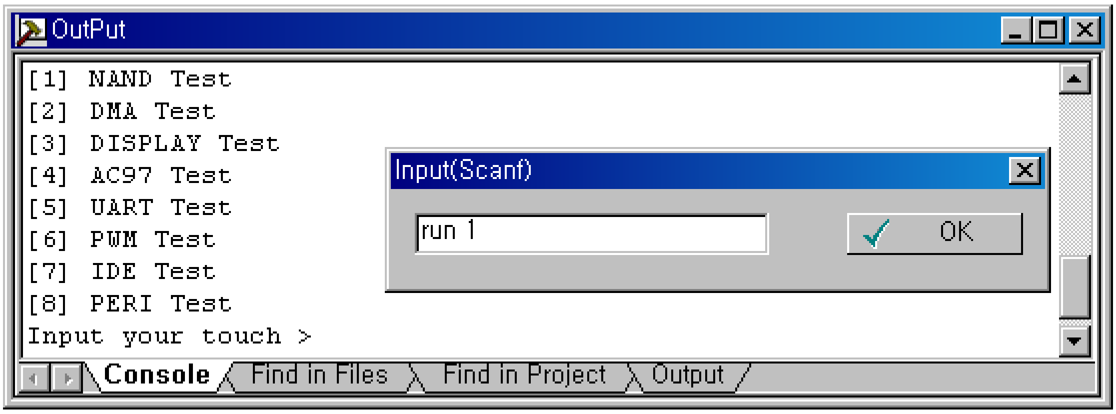


C. Select **File -> Load Image** and load DTK diagnostic program image. (**DTK_diagnostic.axf** or so)

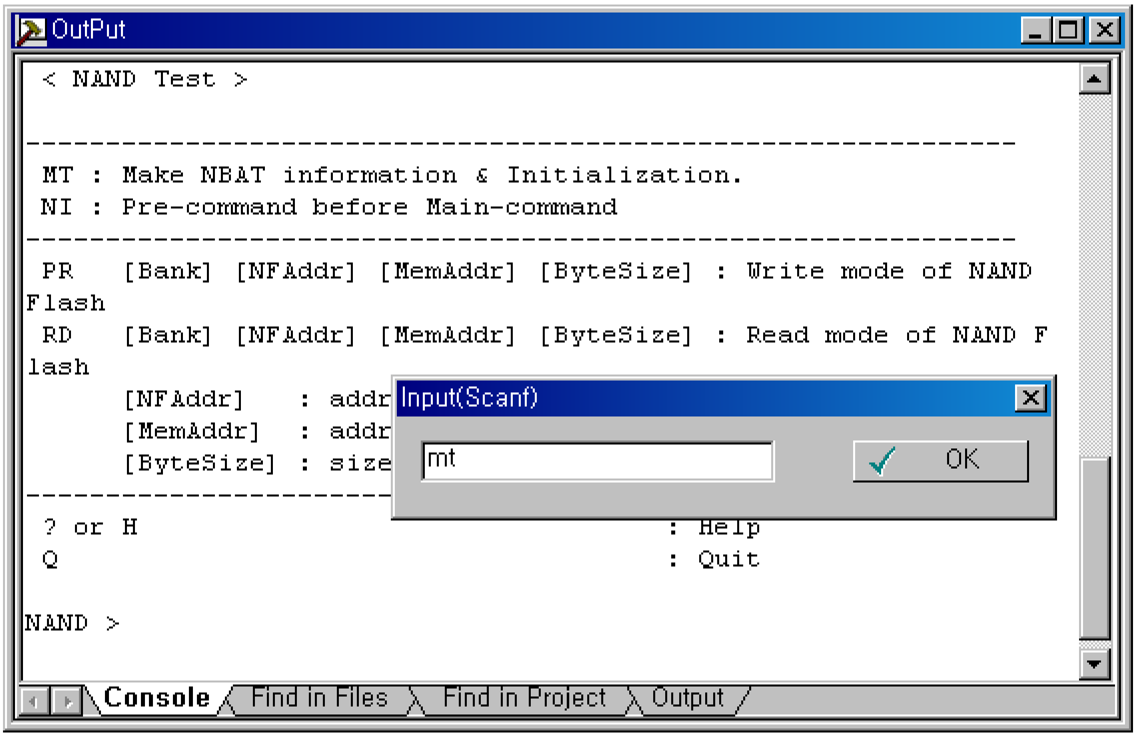
D. Select **Execute -> Go** or Press F5 to run diagnostic program. (You may do this twice to see menu displayed.)

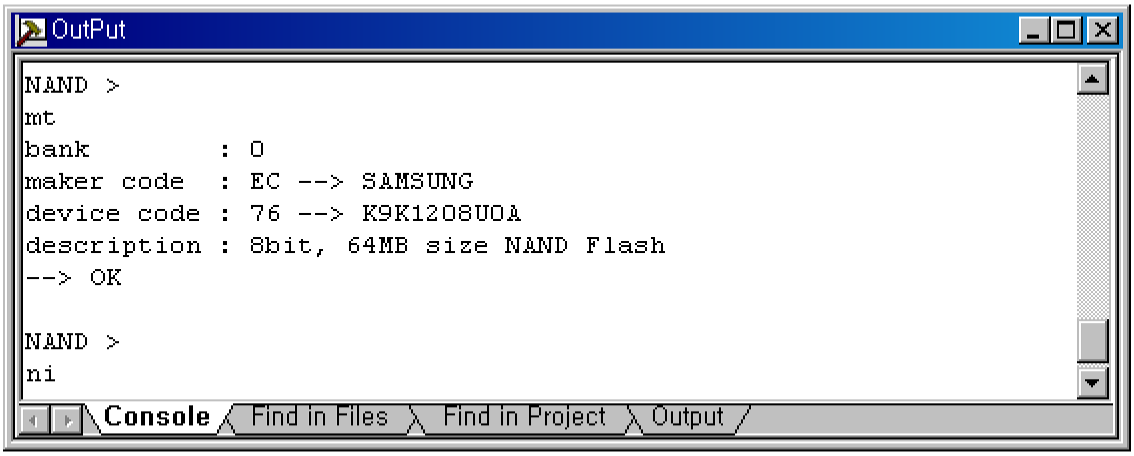


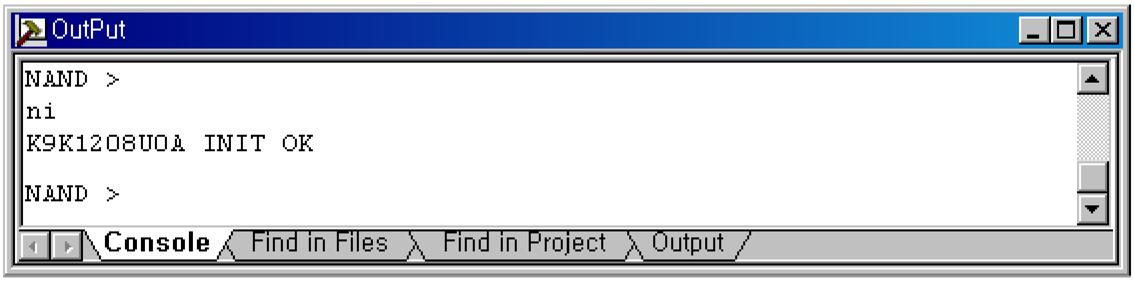
E. Execute “run 1” command in the console window to invoke the “NAND test” mode.



F. Execute “MT” and “NI” command







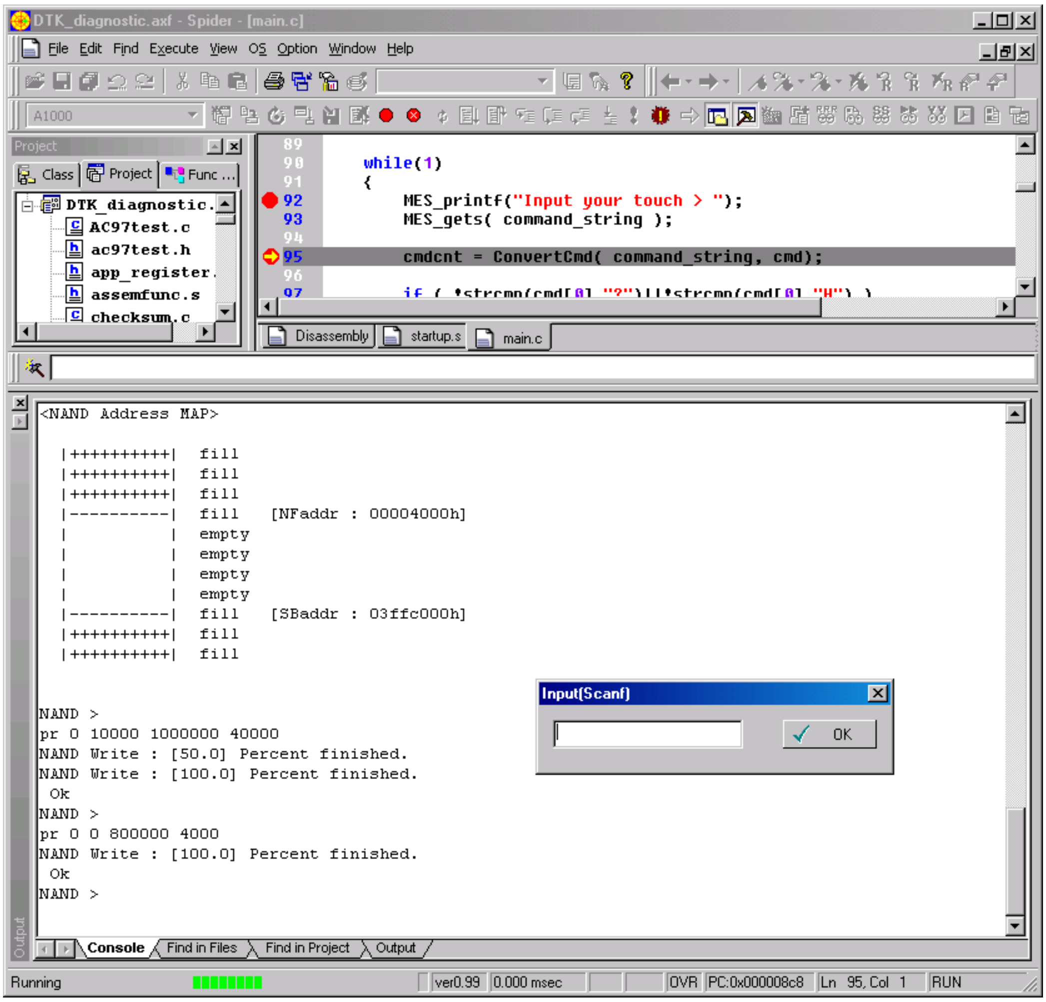
G. Now execute "PR" command to write bootloader images loaded on RAM.

To write Ethernet bootloader image loaded in 1000000h of the RAM, run command like this:

pr 0 10000 1000000 40000

To write bootstrap image loaded in 800000h of the RAM, run command like this:

pr 0 0 800000 4000


DTK_diagnostic.axf - Spider - [main.c]
File Edit Find Execute View OS Option Window Help
Project
Class Project Func ...
DTK_diagnostic.
AC97test.c
ac97test.h
app_register.
assemfunc.s
checksum.c
89
90 while(1)
91 {
92 MES_printf("Input your touch > ");
93 MES_gets(command_string);
94
95 cmdcnt = ConvertCmd(command_string, cmd);
96
Disassembly startup.s main.c
<NAND Address MAP>
|++++++++++| fill
|++++++++++| fill
|++++++++++| fill
|----------| fill [NFaddr : 00004000h]
| | empty
| | empty
| | empty
| | empty
|----------| fill [SBaddr : 03ffc000h]
|++++++++++| fill
|++++++++++| fill
NAND >
pr 0 10000 1000000 40000
NAND Write : [50.0] Percent finished.
NAND Write : [100.0] Percent finished.
Ok
NAND >
pr 0 0 800000 4000
NAND Write : [100.0] Percent finished.
Ok
NAND >
Input(Scanf)
OK
Console Find in Files Find in Project Output
Running ver0.99 0.000 msec OVR PC:0x000008c8 Ln 95, Col 1 RUN

26. Using Debugger with KITL Connection

KITL (Kernel Independent Transport Layer) is a communication layer that manages communication between the development workstation and the target device. You can use KITL for kernel debugging and platform debugging. This BSP supports KITL link with CS8900 Ethernet controller.

Note:

- You should build OS with **Debug** mode for source level debugging.
- Size of Debug mode OS image is much bigger than image with Release mode. Make sure OS image size is less than 30MB. To reduce OS image size, remove large size components such as internet explorer.
- You cannot use KITL with Ethernet and CS8900 Network driver simultaneously. Make sure CS8900 Network driver is not added from catalog.

26.1 Building CS8900 Ethernet Debugging Library

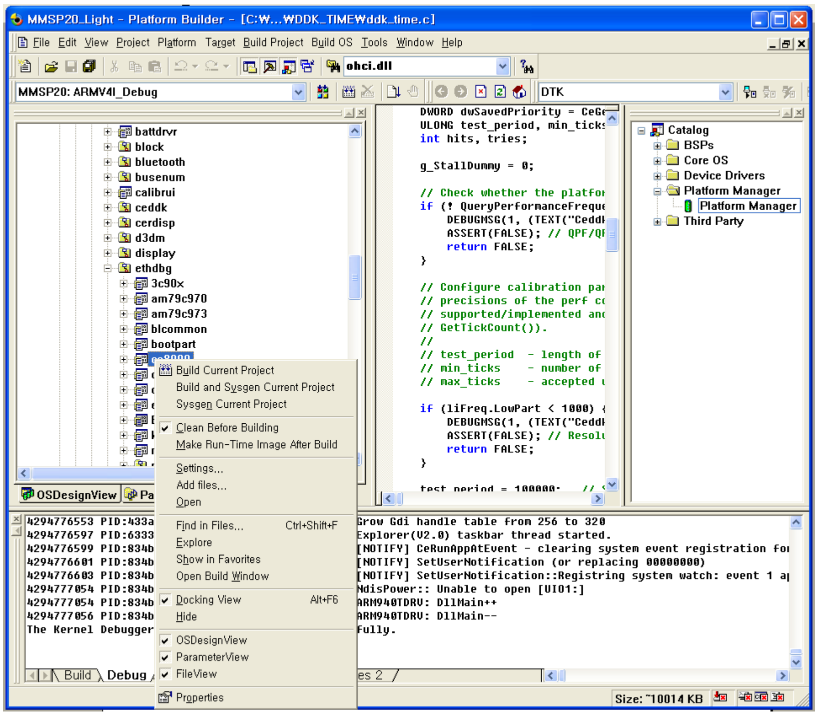
You must modify a source of CS8900 Ethernet debugging library. Location of the source file of this library is **\WINCE500\PUBLIC\COMMON\OAK\DRIVERS\ETHDBG\CS8900**.

Open **cs8900dbg.c** file.

Find the line shown below. Remove this line or cover it with a comment

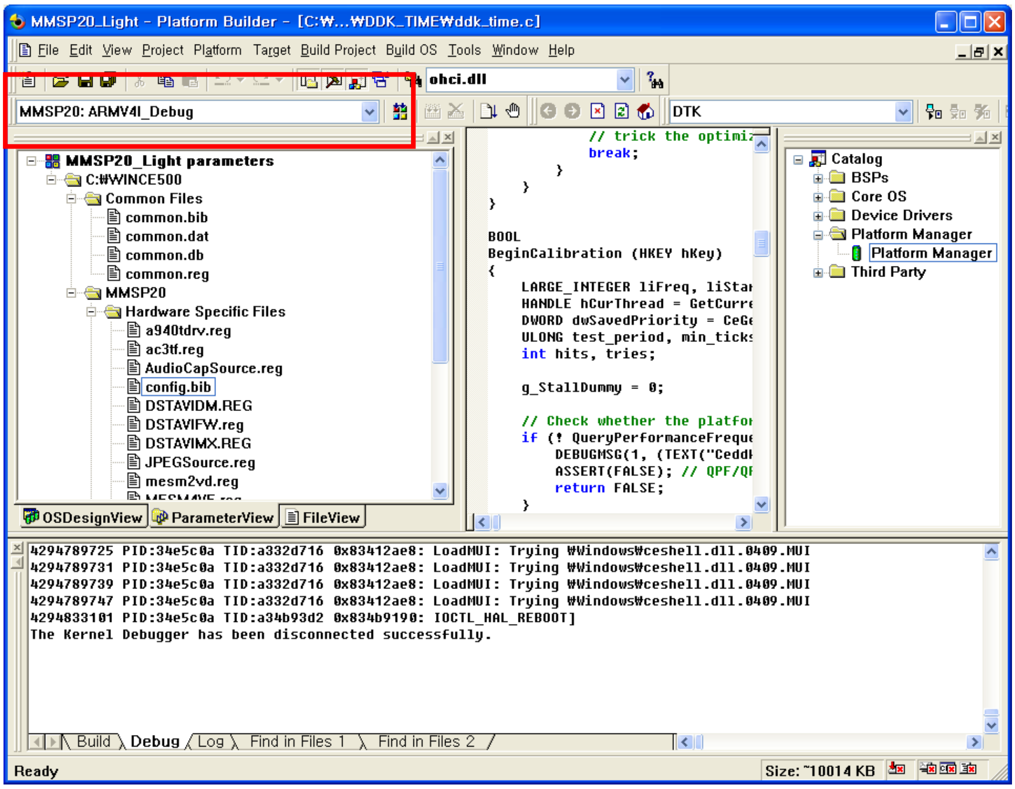
```
#define CS8900_MEM_MODE
```

Build this library. Select CS8900 directory on **FileView** and click it with right button of the mouse. Then perform building by selecting **Build Current Project**. (For detailed information of building part of source code, see **Building WinCE OS** -> **Building Part of BSP Source** chapter of this document.)



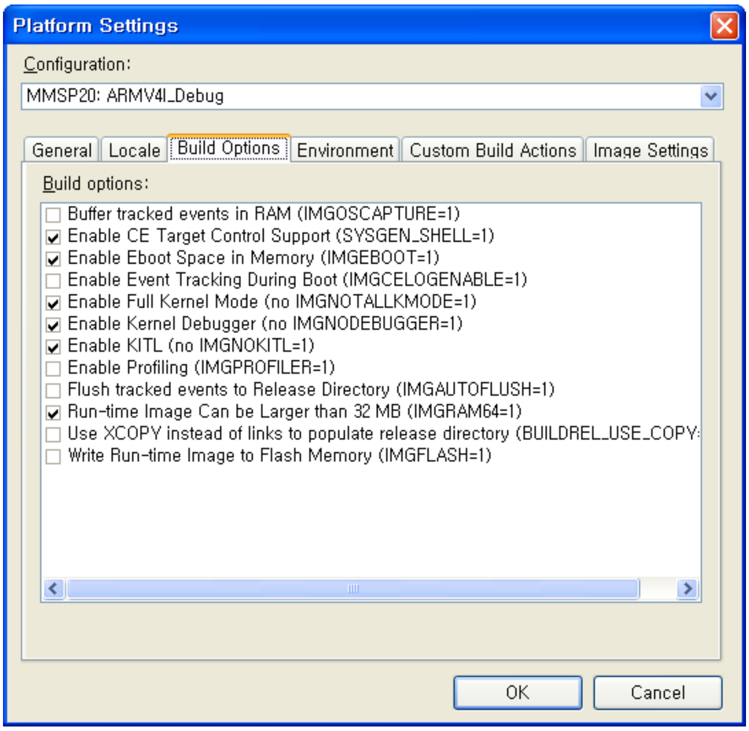
26.2 Setting and Building OS Image for KITL Link

For source level debugging, OS image should be built Debug mode.

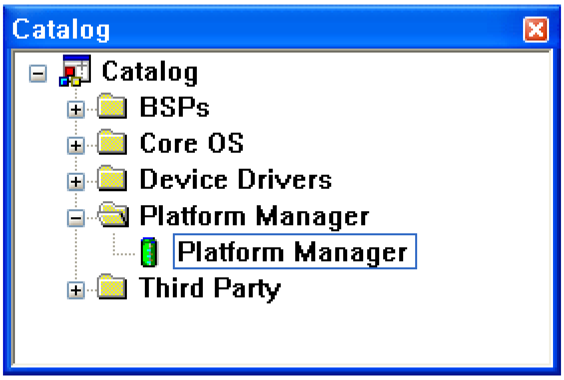


Select **Platform** -> **Settings** menu and select **Build Options** tab.

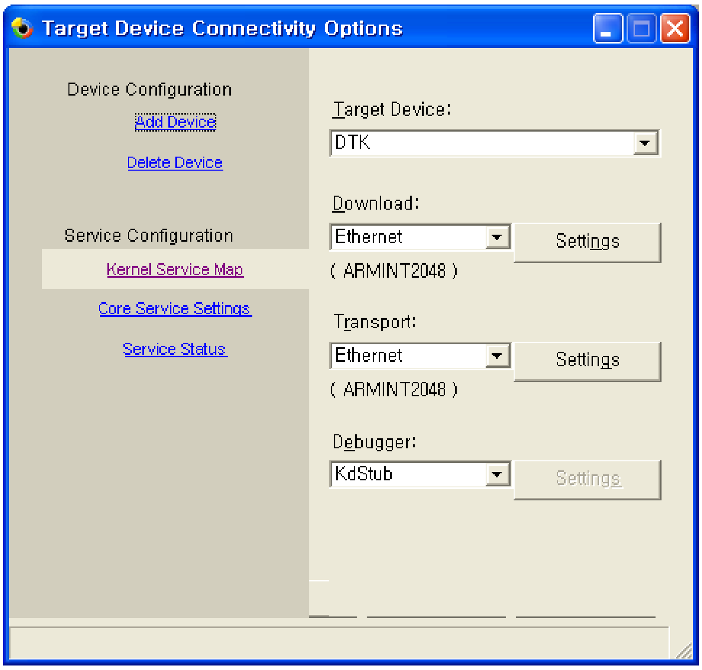
Select **Enable Kernel Debugger** and **Enable KITL** options.



Add **Platform Manager** -> **Platform Manager** from **Catalog** list.



Select **Target** -> **Connectivity Options** and select **Debugger** of current connectivity selection as **Ethernet**.

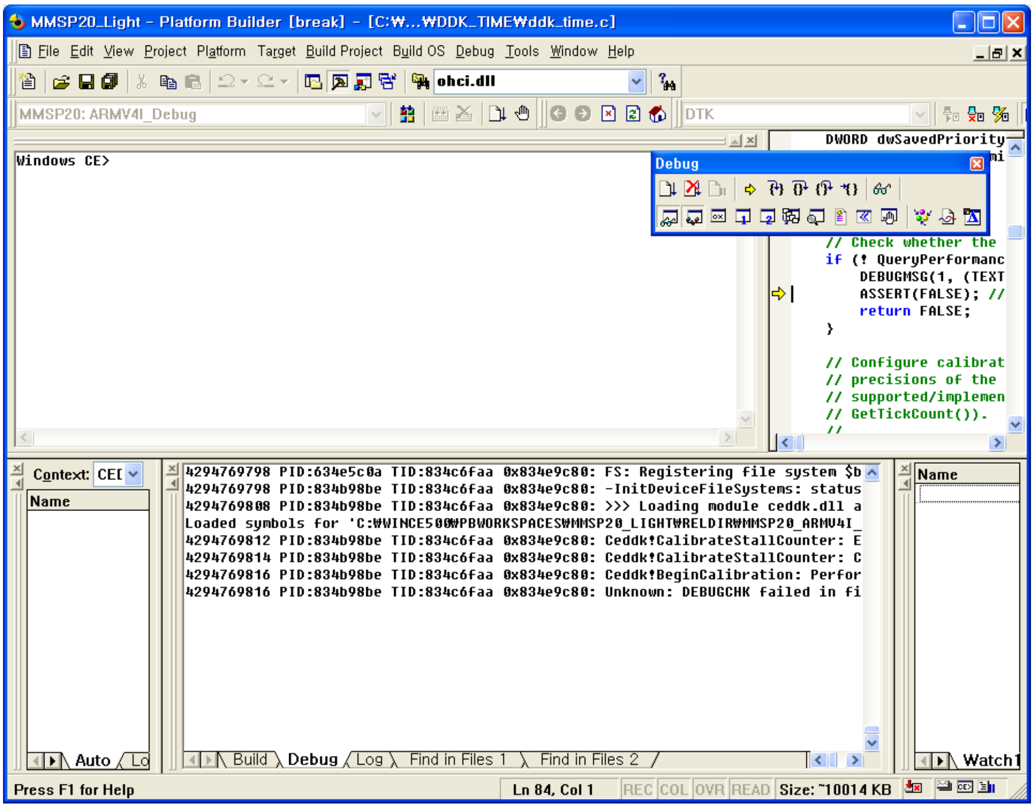


Build OS by selecting **Build OS** -> **Sysgen**.

26.3 Connect Platform Builder and Target System with KITL

Download OS image with Ethernet bootloader.

After download is completed, you can see this debugging window. Now you can use kernel debugging.


MMSP20_Light - Platform Builder [break] - [C:₩...₩DDK_TIME₩ddk_time.c]
File Edit View Project Platform Target Build Project Build OS Debug Tools Window Help
ohci.dll
MMSP20: ARMV4I_Debug
DTK
Windows CE>
Debug
DWORD dwSavedPriority
// Check whether the
if (! QueryPerformanc
DEBUGMSG(1, (TEXT
ASSERT(FALSE); //
return FALSE;
}
// Configure calibrat
// precisions of the
// supported/implemen
// GetTickCount()).
Context: CEI
Name
4294769798 PID:634e5c0a TID:834c6faa 0x834e9c80: FS: Registering file system $b
4294769798 PID:834b98be TID:834c6faa 0x834e9c80: -InitDeviceFileSystems: status
4294769808 PID:834b98be TID:834c6faa 0x834e9c80: >>> Loading module ceddk.dll a
Loaded symbols for 'C:₩WINCE500₩PBWORKSPACES₩MMSP20_LIGHT₩RELDIR₩MMSP20_ARMV4I_
4294769812 PID:834b98be TID:834c6faa 0x834e9c80: Ceddk!CalibrateStallCounter: E
4294769814 PID:834b98be TID:834c6faa 0x834e9c80: Ceddk!CalibrateStallCounter: C
4294769816 PID:834b98be TID:834c6faa 0x834e9c80: Ceddk!BeginCalibration: Perfor
4294769816 PID:834b98be TID:834c6faa 0x834e9c80: Unknown: DEBUGCHK failed in fi
Auto
Build
Debug
Log
Find in Files 1
Find in Files 2
Name
Watch1
Press F1 for Help
Ln 84, Col 1
REC COL OVR READ
Size: ~10014 KB

27. Using Debugging Tools with TCP/IP Networking

Debugging application developed in eMbedded Visual C++ or using other remote debugging tools are possible with TCP/IP transport connection to DTK board.

Available remote tools are:

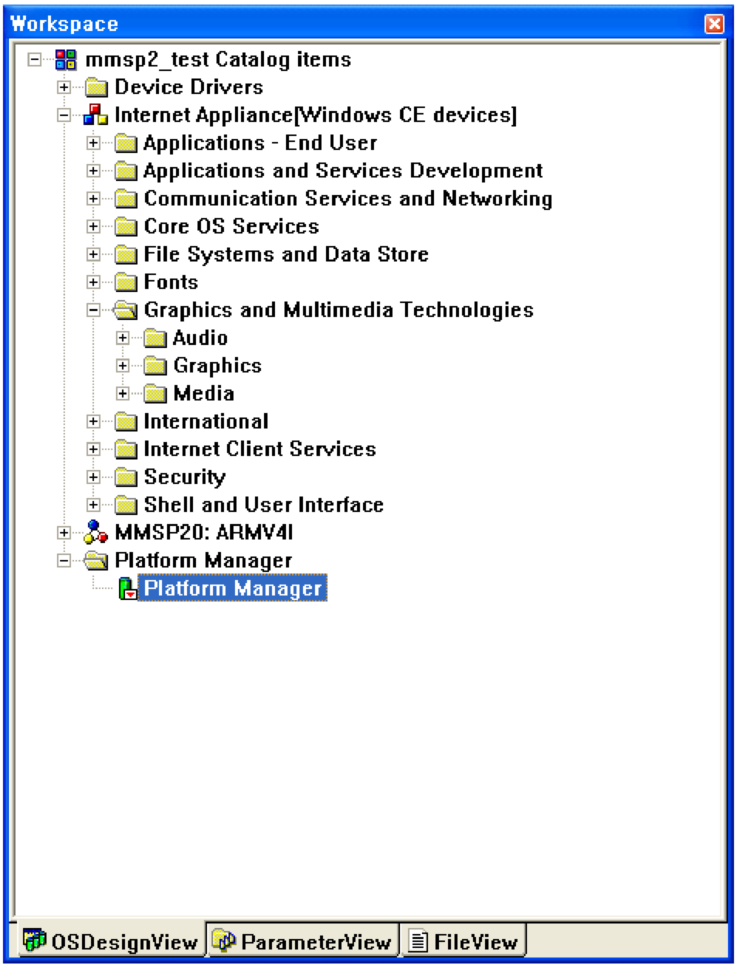
- Remote Call Profiler
- Remote File Viewer
- Remote Heap Walker
- Remote Kernel Tracker
- Remote Performance Monitor
- Remote Process Viewer
- Remote Registry Editor
- Remote SPY
- Remote System Information
- Remote Zoom-in

Local Area Network driver should be set correctly to use these tools.

27.1 Required Components

Add components needed for connection to target board.

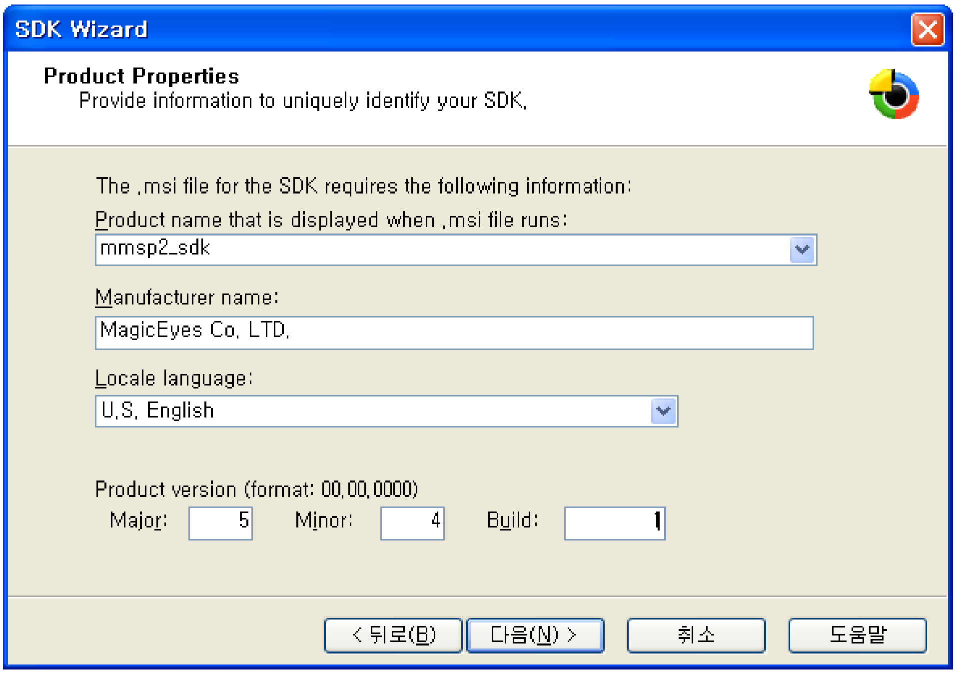
- **Platform Manager** → **Platform Manager**

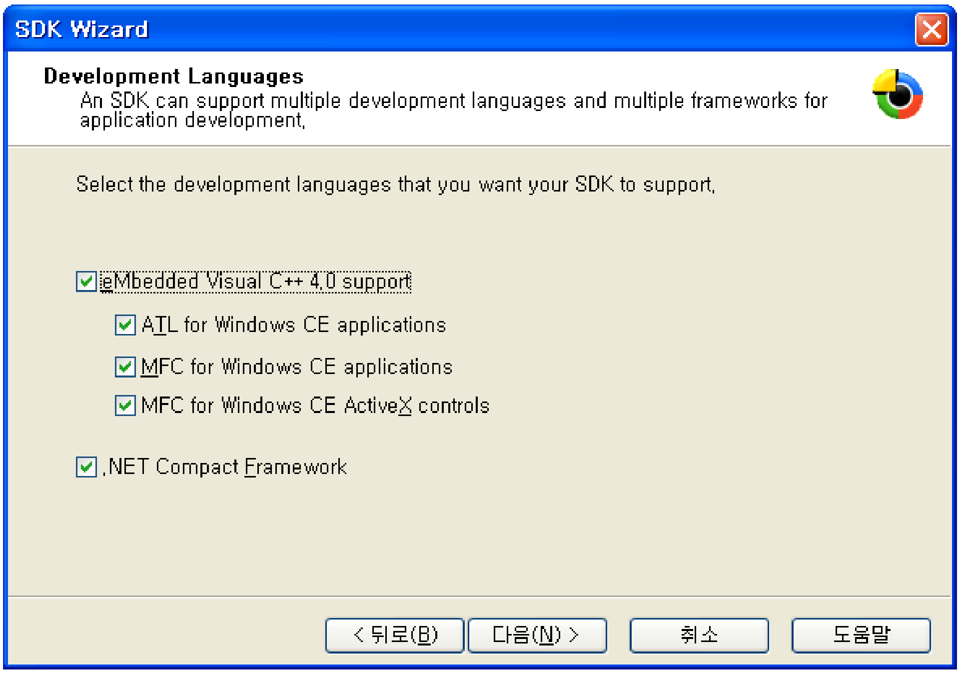


After add all required components, you must rebuild platform to apply new setting.

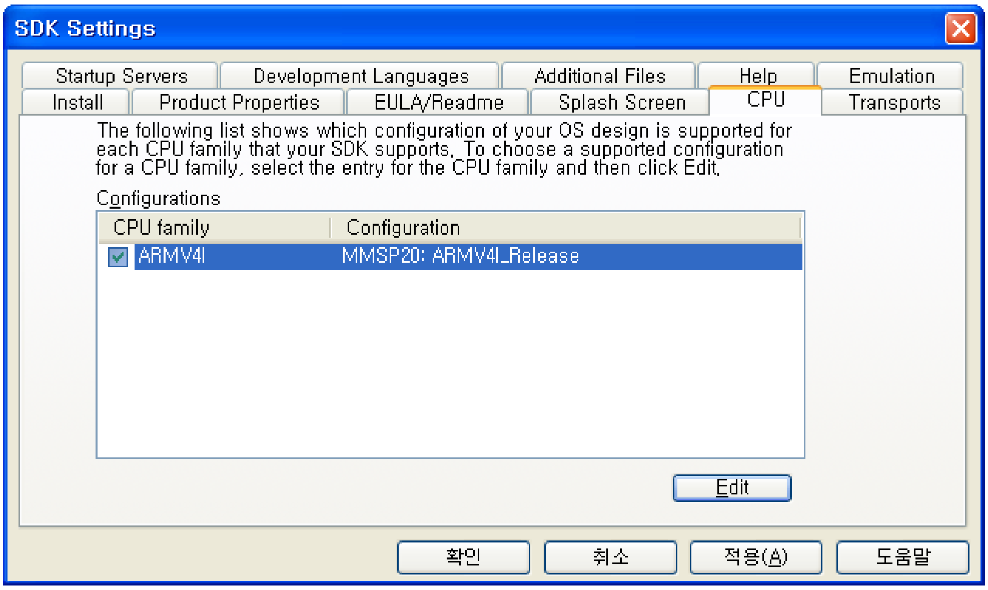
27.2 Building and Installing SDK for Target Connection

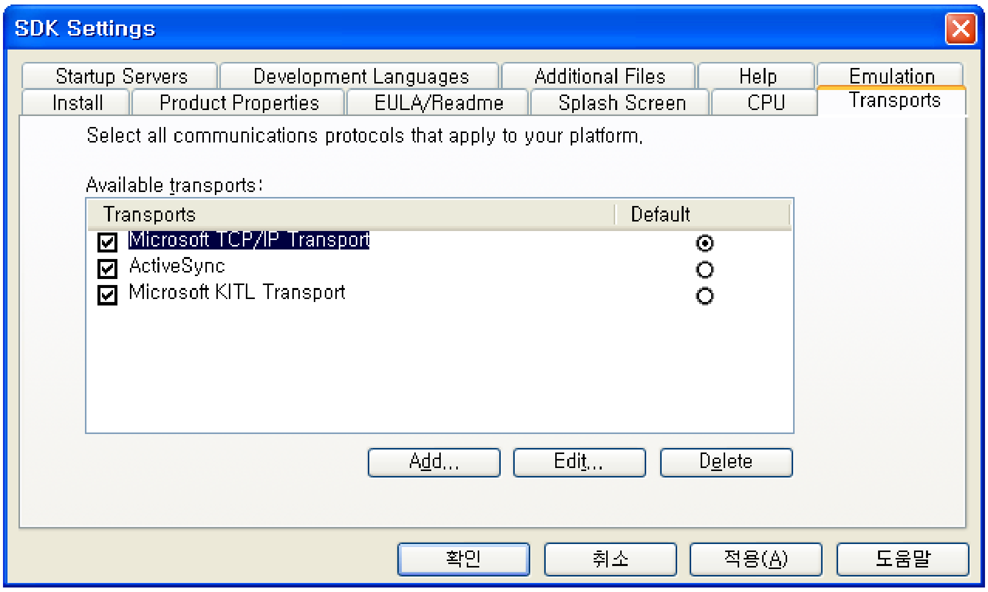
Select **Platform** → **SDK** → **New SDK** menu and make a new SDK as shown below.

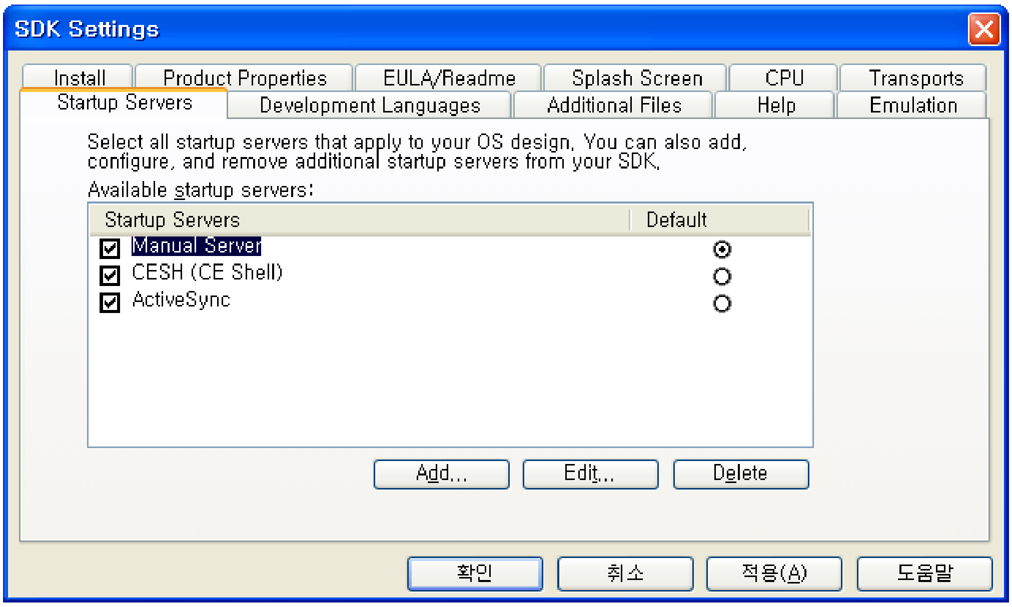

SDK Wizard
Product Properties
Provide information to uniquely identify your SDK.
The .msi file for the SDK requires the following information:
Product name that is displayed when .msi file runs:
mmsp2_sdk
Manufacturer name:
MagicEyes Co. LTD.
Locale language:
U.S. English
Product version (format: 00.00.0000)
Major: 5
Minor: 4
Build: 1
< 뒤로(B)
다음(N) >
취소
도움말


SDK Wizard
Development Languages
An SDK can support multiple development languages and multiple frameworks for application development.
Select the development languages that you want your SDK to support.
eMbedded Visual C++ 4.0 support
ATL for Windows CE applications
MFC for Windows CE applications
MFC for Windows CE ActiveX controls
.NET Compact Framework
< 뒤로(B)
다음(N) >
취소
도움말

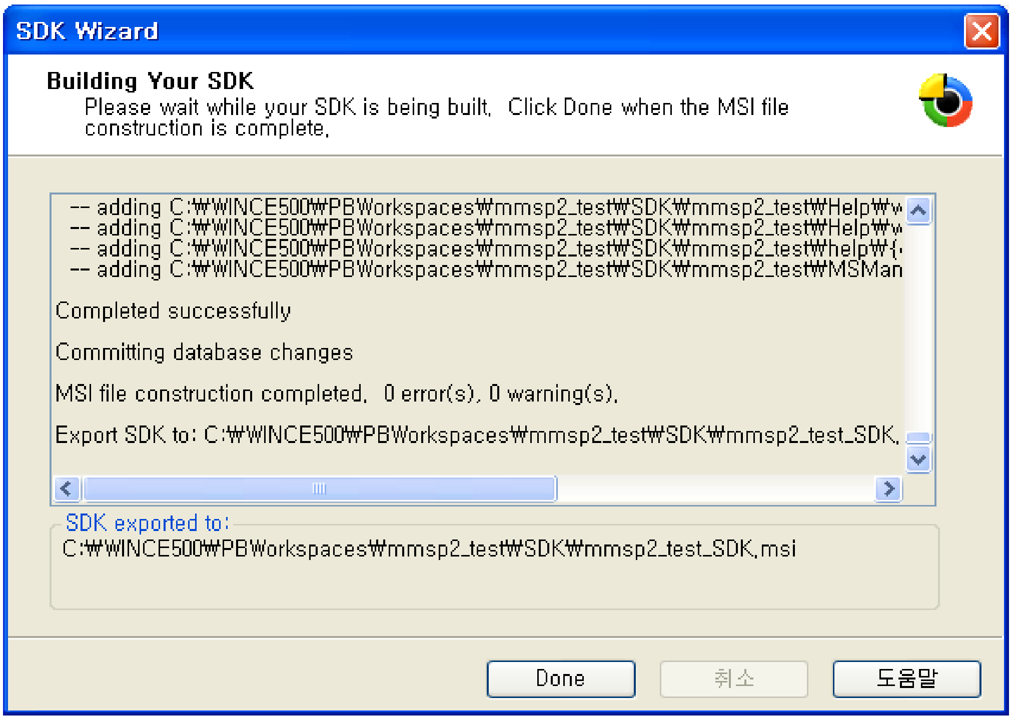
Select **Platform** → **SDK** → **Configure SDK** menu and make a new SDK as shown below.







After setting is completed, select **Platform** → **SDK** → **Build SDK** and build SDK.

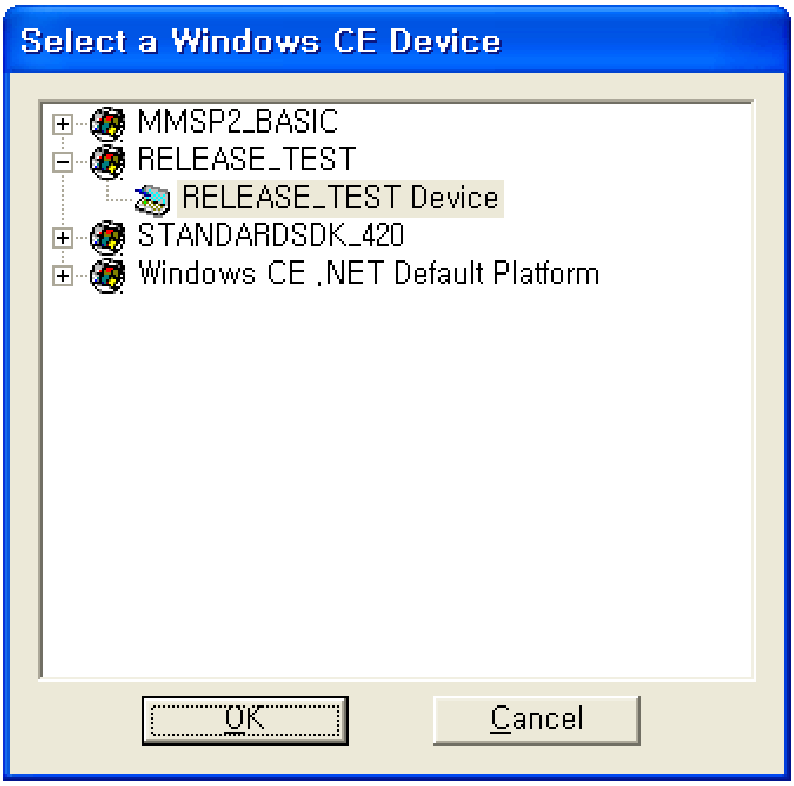


Go to folder that has built SDK and run .msi to install SDK.

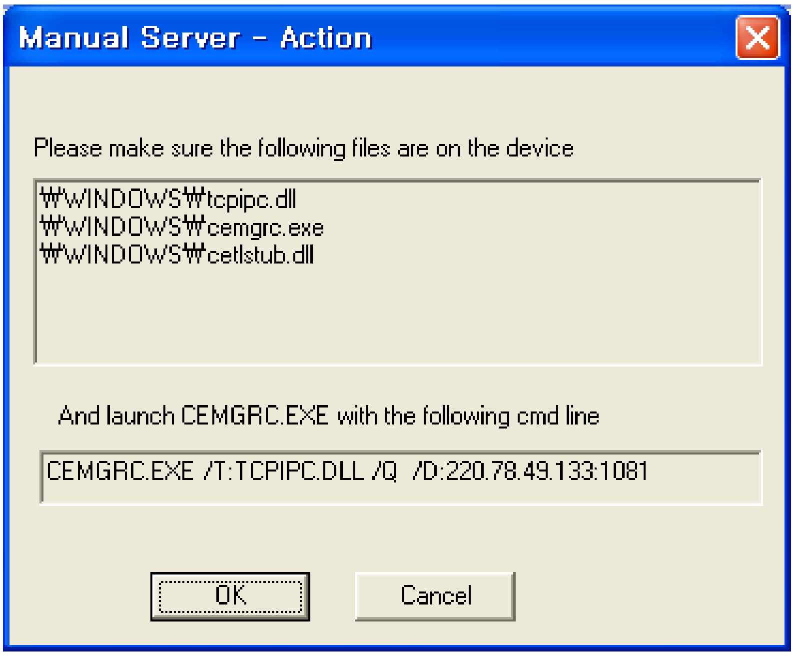
* Reboot system to apply installed SDK.

27.3 Connecting to Target by Using Debugging Tools

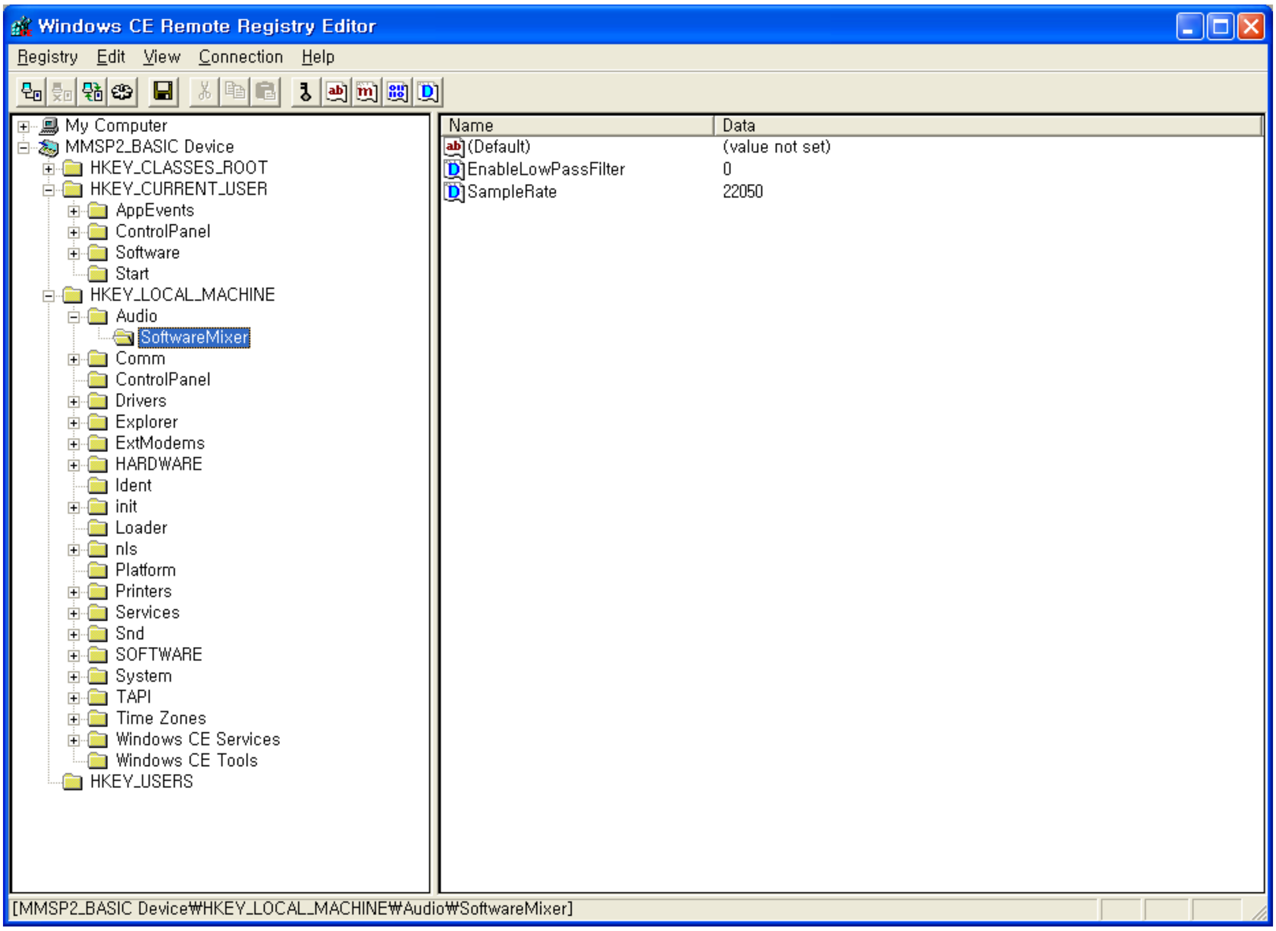
Select device of newly installed SDK when selection window is shown.



Select Start -> Run by pressing start button in DTK board and run as shown.



Now eMbedded C++ debugger or remote tools can be used. This is an example of running Remote Registry Editor.



28. Applying Customized Board Configuration

28.1 About IOManager

When you develop your own MMSP2 based board, you should modify BSP source according to your own board configuration. MMSP2 BSP for Windows CE 5.0 provides IOManager that contains common settings of GPIO(General Purpose IO), DMA, etc.

In the old version of the BSP source, kernel and drivers had their own GPIO or DMA configuration and handling functions. With old version of BSP source, you must search all BSP source file and modify each source file that contains board specific configuration. But with IOManager, you can apply new configuration of GPIO or DMA by modifying one source file of the library source.

28.2 Source Code of IOManager

Source of IOManager is in PROTO\IOManager. You should build IOManager folder first if you want to apply modification with partial building.

PROTO\IOManager

Main.c – Contains initialization / de-initialization functions

GPIOPinDefine.c – Contains GPIO pin configuration.

GPIOManager.c – Contains abstracted functions for GPIO handling.

PROTO\INCLUDE_PROTO

IOManager.h – Contains declaration of initialization / de-initialization functions and abstracted functions.

GPIOManager.h – Contains declaration of pin configuration and GPIO handling function array.

KERNEL\HAL

int_gpio.c – This is a part of kernel source and not a part of IOManager. It requires IOManager library.

28.3 Modifying GPIO Settings

28.3.1 GPIO Pin Configuration

GPIO Pin configuration is in **GPIOPinDefine.c** file.

A pin definition is a combination of "GROUP" and "PIN Number." (For more information of group and pin number, see "GENERAL PURPOSE I/O (GPIO)" of MP2520F data book.)

Pin definition of IOManager is a structure type shown below:

```
typedef struct _ioman_gpio_signal {
    DWORD group;
    USHORT pin;
} IOMan_GPIO_Signal;
```

In the DTK 4th or later board, IRQ signal of the external Ethernet controller is connected to GPIO group N 4. This is defined as:

```
IOMan_GPIO_Signal ETHER_IRQ                    = {GroupN, 4};
```

28.3.2 Modifying Existing GPIO Configuration

Open GPIOPinDefine.c and modify pin definition.
For example, if your board uses GPIO I 15 as SD_CD signal instead of GPIO I 14, find definition of SD_CD and modify the pin number.

Modify this code

```
IOMan_GPIO_Signal SD_CD                        = {GroupI, 14};
```

as

```
IOMan_GPIO_Signal SD_CD                        = {GroupI, 15};
```

After modifying source code, build all BSP. If you want reduce build time, you can build library and other parts of BSP one by one.

1. Build PROTO\IOManager.
2. Build KERNEL.
3. Build the driver that uses modified GPIO signal.

28.3.3 Adding New GPIO Configuration

To add new GPIO signal definition, you must modify both **GPIOPinDefine.c** and **GPIOManager.h**.
At first, add new pin definition to **GPIOPinDefine.c** file.

For example, if you want to add a new GPIO pin definition of GPIO O 0 and name it as GPIO_POO, add code shown below:

```
IOMan_GPIO_Signal GPIO_POO                      = {GroupO, 0};
```

Then add "extern" declaration of this definition to **GPIOManager.h** file.

```
extern IOMan_GPIO_Signal GPIO_POO;
```

To apply this modification, build IOManager and other related parts of BSP.

28.4 GPIO Handling Functions of IOManager

CAUTION: Guidance of this chapter is needed only when you develop a new driver or modify major part of existing drivers. When your board has different GPIO pin mapping, it is recommended that you modify just GPIO pin configuration rather than GPIO handling functions.

28.4.1 GPIO Function Array

IOManager has GPIO function array which wraps functions of proto GPIO library. (GPIOLIB.lib) When this function array is used with GPIO pin configuration, you don't have to modify function name when GPIO pin configuration is changed.

For example, if you want add a function that sets output value of CF0_RESET as 1:

```
    pfIOMan_GPIO_SetGPIOOut[CF0_RESET.group](CF0_RESET.pin, 1);
```

Function array mappings are in **GPIOManager.c** file.

28.4.2 Abstracted GPIO Handling Functions

GPIOManager.c contains abstracted GPIO handling functions which are used by drivers. For example, for SD/MMC card detection, SD/MMC loader driver uses IOMan_SD_CD_GetValue() function in GPIOMananger.c.

```
BOOL IOMan_SD_CD_GetValue(void)
{
  BOOL ReturnVal;
```

```
    WAIT_GPIO_FLAG();
    ReturnVal = (pfIOMan_GPIO_GetGPIOIn[SD_CD.group]() & _BIT(SD_CD.pin)) >>
SD_CD.pin;
    RELEASE_GPIO_FLAG();

    return ReturnVal;
}
```

28.4.3 Using Access Control for GPIO Handling

To prevent confliction or corruption caused by access of other threads, IOManager provides access control macro. All GPIO access code should be placed between **WAIT_GPIO_FLAG** and **RELEASE_GPIO_FLAG**.

```
VOID IOMan_SIGNAL_SetMode(void)
{
    WAIT_GPIO_FLAG();
    // Place GPIO access function here
    RELEASE_GPIO_FLAG();
}
```

28.4.4 Adding an Abstracted GPIO Handling Function to IOManager

You can add abstracted handling function of GPIO pin to the IOManager and call it on your driver. To add a function,

- Write new function body to GPIOManager.c
- Write declaration of new function to IOMananger.h
- Build IOManager

In the abstracted function, you must use predefined GPIO pin mapping and GPIO function array for GPIO handling.

Use **WAIT_GPIO_FLAG** and **RELEASE_GPIO_FLAG** macro for access control. (See previous chapter)

28.5 Modifying DMA Settings

Iomanager.h file contains DMA channel number. All DMA channel allocation should be placed here. And all driver should use channel number defines in this header file.

28.6 Linking IOManager Library

NOTE: Guidance of this chapter is needed when you develop a new driver.

If you want to write a driver which controls GPIO signal, link IOManager and use GPIO pin configuration in the library.

To link IOManager to a driver, modify **SOURCES** file and add a line shown as bold:

```
TARGETLIBS=
$(_TARGETPLATROOT)\lib\$(_CPUINDPATH)\IOMANAGER.lib   \
```

* CAUTION: Do not specify GPIOLIB.lib on TARGETLIBS list.

Add **PROTO\INCLUDE_PROTO** as a include path.

```
INCLUDES= $(INCLUDES) ; ..\..\PROTO\INCLUDE_PROTO
```

(beware of relative path)

Then you must add initialization and de-initialization functions to your own driver.

Call **IOMan_Initialize()** function on the initialization function of the new driver.

Call **IOMan_Deinitialize()** function on the de-initialization function of the new driver.